ROYAL CANADIAN NAVY
CASUALTY CLEARING PROCEDURES MANUAL

OPI: Commanders Sea Training Pacific/Atlantic

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ROYAL CANADIAN NAVY
CASUALTY CLEARING (CC) PROCEDURES MANUAL

PREFACE

1. The Royal Canadian Navy's Casualty Clearing (CC) Procedures Manual is effective upon receipt. It is promulgated under the authority of the Commander Sea Training Group, who is responsible to Commander MARLANT for pan RCN direction and management of all readiness policies, documentation and processes supporting operational readiness. The Casualty Clearing (CC) Procedures Manual is issued in electronic format only and is available on the Sea Training Group DWAN homepage.

2. The Casualty Clearing (CC) Procedures Manual is the first authoritative tactical level doctrine on the Casualty Clearing (CC) Procedures roles and responsibilities within the Internal Battle. Casualty Clearing (CC) procedures and best practices are also addressed.

3. This long-awaited doctrine is linked to Joint Canadian Armed Forces doctrines as well as the RCN's Readiness and Sustainment Policy (CFCD 129). It provides guidance on the training and education of Medical Teams and Casualty Clearers who will be employed in the RCN into the future. It follows in the footsteps of well-established Damage Control policies and procedures, and is consistent with long standing naval Casualty management doctrines within NATO, across the US Armed Forces as well as within Her Majesty's Armed Forces.

4. Units are to review the Casualty Clearing (CC) Procedures Manual, implement its content and, where deemed necessary, propose amendments to coastal Sea Training organizations.

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LIST OF ACRONYMS

AFT CC	After Casualty Clearing
C2	Command and Control
Cbt O	Combat Officer
CC HQ	Casualty Clearing Headquarters
CC Org	Casualty Clearing Organization
CCT	Casualty Clearing Team
CDC	Casualty Data Coordinator
CFJP	Canadian Forces Joint Publication
CST(A)	Commander Sea Training (Atlantic)
CST(P)	Commander Sea Training (Pacific)
CSTG	Commander Sea Training Group
D/MSEO	Deputy MSEO
DC	Damage Control
DCI	Damage Control Instruction
DCO	Damage Control Officer
DIO	Damage Information Officer
ERT	Emergency Response Team
ERT Ops	Emergency Response Team (Operations Room)
FFE	Fire Fighting Equipment
Fleet PA	Fleet Physician Assistant
FSMO	Fleet Support MO
FWD CC	Forward Casualty Clearing
HT	Hull Technician
I/C	Person In Charge
MED HQ	Medical Headquarters
MMN	Meet Me Net
NATO	North Atlantic Treaty Organization
ORO	Operations Room Officer
PPE	Personal Protective Equipment
SMA	Senior Medical Authority (Ship or Boat MO/PA/Med Tech)
SSO	Ship’s Standing Orders
ST PA	Sea Training Physician Assistant
XO	Executive Officer

CHAPTER 1 – INTRODUCTION

101. General

1. The Casualty Clearing (CC) Procedures Manual is the first authoritative tactical level doctrine on the Casualty Clearing Procedures roles and responsibilities within the Internal Battle. Casualty Clearing procedures and best practices are also addressed.

2. The Casualty Clearing Procedures Manual is to be viewed as the authoritative reference for RCN Casualty Clearing Procedures operations with the exception of aspects of the provision of individual medical care, which will be covered by Specific HS Policies and Procedures and promulgated through the RCN Surgeon.

102. Applicability and Responsibilities

1. The CC Procedures Manual is effective upon receipt and is applicable to all RCN Vessels with minor class modifications as indicated where applicable. It is promulgated under the authority of the Commander Sea Training Group, who is responsible for pan RCN direction and management of all readiness policies, documentation and processes supporting operational readiness.

2. As a minimum, this publication along with the relevant Ship's Standing Orders (SSO) articles, DC Manual sections, and applicable Damage Control Instructions (DCI), are to be read and implemented by the ships' Casualty Clearing Organizations.

3. The procedures and doctrine in this manual apply to training, wartime operations and military operations other than war.

4. The OPI for this publication is Commander Sea Training Group.

103. Amendments

All amendment proposals are to be forwarded to the Coastal Sea Training staff for furtherance to the STG OPI.

104. Related Publications

1. The following list of publications include direction or guidance with respect to Casualty Clearing and should be read in conjunction with this manual:

   a. Ship's Standing Orders Vol 1 and Platform Specific Vol 2 sections, which include specifics on:

      (1) General alarms;
      (2) Emergency in Harbour;
      (3) Emergencies at sea;

(4) Action/Emergency Org at sea; and
(5) key personnel closing up in emergencies and at action; and

b. C-23-045-000/AG-001 Damage Control (Sea) Vol 1 Shipboard Damage Control Change 1 (17 July 2009).

CHAPTER 2 –CASUALTY CLEARING TEAM (CCT) DOCTRINE

201. Introduction

1. In accordance with CFJP 01 Canadian Military Doctrine (B-GJ-005-000/FP-001), "A sound doctrinal framework provides the basis for operations and training, guides Commanders and helps individuals to think more clearly in the fog of war." Further, the doctrine notes that a common approach based upon methodical thinking and supported by effective training can lead to "consistent behaviour, mutual confidence and effective collective action. Well-developed military doctrine is inherently flexible, allowing Commanders to seize the initiative and adopt unorthodox or imaginative courses of action as opportunities arise in the "battlespace.""

2. CAF doctrine encompasses all three levels of military activities: strategic, operational, and tactical. Environment-specific doctrine is generally at the tactical level.[1] As CFJP 01 indicates "tactical level doctrine is concerned with planning and directing military resources in battles, engagements and/or activities within a sequence of major operations to achieve operational objectives." For the purpose of providing the fundamental principles by which CCT will guide its actions in support of objectives, a tactical doctrine is therefore necessary.

202. A Common Philosophy

1. As outlined in CFJP 01 Canadian Military Doctrine, the CAF has adopted six guiding principles applicable to all environments, one of which is directly applicable to CCT operations: Mission Command.

2. Mission Command is the CAF philosophy of Command which articulates the dynamic and decentralized execution of operations guided throughout by a clear articulation and understanding of the overriding Commander's intent. This philosophy demands the aggressive use of initiative at every level, a high degree of comfort in ambiguity, and a tolerance for honest failure.[2] Mission Command has its origins in the German Army concept of Auftragstaktik, and is often contrasted with a Command style reliant on procedural direction and control.[3]

3. The CAF Mission Command philosophy has five fundamentals[4], which must inform the CC Org tactical level doctrine's fundamental principles:

a. Unity of Effort. Subordinates who understand the intent of their immediate superiors and those two levels up enhance unity of effort. This allows

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[1] A-GJ-025-0A1/FP-001 CFJP A1 *Doctrine Development Manual*, issued under the authority of the Chief of Defence Staff, January 2010
[2] A-AP-005-000/AP-001 *Duty with Honour – The Profession of Arms in Canada*
[3] A-PA-005-000/AP-004 *Leadership in the Canadian Forces – Conceptual Foundation*, published by the CDA Canadian Forces Leadership Institute, 2005
[4] B-GL-300-003/FP-000 *Command*, issued under the authority of the Chief of Defence Staff, 21 July 1996

subordinates to be aware of the 'big picture' and makes it far more likely that they will continue to act purposefully in an unexpected situation;

b. Decentralizing Authority. Decision thresholds should be set as low as possible. This sets the conditions for making appropriate decisions swiftly in the confusion and uncertainty of battle. It also reduces the need for passing all but essential information up and down the chain of command. The more fluid the circumstances, the lower the decision threshold should be set. Devolution of decision-making requires delegation of specific authorities;

c. Trust. For Mission Command to function effectively, a superior needs to trust, and earn the trust, of his/her subordinates;

d. Mutual Understanding. A professional understanding of doctrine, drills and procedures, including the language of command, is held in common. Commander's intentions must be clear to subordinates if they are to understand what they are to achieve; and

e. Timely and Effective Decision-Making. Battles may be lost, or victory delayed, because of a Commander's failure to recognize favourable circumstances and make sound and timely decisions.

203. CCT Tactical Doctrine

1. The CCT Fundamental Principles, informed by Mission Command philosophy in their application, shall guide the establishment of the CCT within the Internal Battle Organization on current and future platforms. These principles shall guide the planning and conduct of CCT operations or training as it relates to the management of casualties within the RCN.

2. It is the Ships Senior Medical Authority's (SMA) leadership responsibility to apply this doctrine to the ship's CCT organization and use judgment in its application. The SMA must understand that the content of this manual is only prescriptive as long as it fulfills his or her Commanding Officer's Intent, and that any initiatives to improve the alignment of the CCT organization, procedures and practices to this intent must be pursued within the bounds of the above mentioned doctrine.

3. The fundamental keystones of all medical care onboard vessels are to:

a. manage the emergency phase of care if it exists;

b. stabilize the patient enough to consult with shore based medical experts;

c. further implement recommendations received from those shore based experts;

d. decide if the person can be/must be evacuated immediately or not, in consultation ships Command and OPs staff;

e. further stabilize and prepare to hold until evac resources are available, or prepare for immediate evacuation; and

f. Hand over care to the receiving physician at the hospital you are evacuating the patient to, and inform the CF H Svc CoC that an evacuation has taken place and the details of the patient status on departure.

4. No casualty incident is complete until all of the above steps have taken place, regardless of the integral medical support available on the vessel. In addition to the above, records, witness statements and all evidence related to the incident must be preserved to the maximum extent possible in order to capture lessons learned and to provide for investigative procedures if required. Standard medical confidentiality rules still will apply and advice shall be sought from the applicable fleet support MO when an investigation is ordered.

CHAPTER 3 – ROLES, RESPONSIBILITIES AND EXPECTATIONS OF THE CCT

301. General

1. This chapter will discuss the basic roles and responsibilities of the CCT, the typical CCT organization, and the expectations of CCT personnel.

2. As indicated in Section 203, the SMA must understand that the content of this manual is only prescriptive as long as it fulfills his or her Commanding Officer's Intent. Any initiatives to improve the alignment of the CCT organization, procedures and practices to this intent must be pursued within the bounds of the doctrine outlined in Chapter 2 and Ship's Standing Orders.

302. Basic Roles and Responsibilities

1. In general terms, the CCT organization must be able to quickly identify and respond to casualties, provide effective first aid, and assist the SMA in the provision of care and evacuation of casualties. To achieve this, there are four basic roles and responsibilities assigned to the CCT:

a. Support to the DC effort. This relates to the set of responsibilities with respect to damage control incidents like fires and floods, including rapid response, setting boundaries, and assisting the DCO in developing a common operational picture of damage control incidents within the ship. Supporting the DC effort also prevents the spread of fires/floods, preserving or rapidly restoring combat capabilities, providing "Buddy" and "Self" First Aid;

b. Identify and Assess Casualties. This relates to making immediate assessments of the location of potential casualties and assessing their status and reporting their condition to command;

c. Maintenance of unit First Aid stats. The Ships CC Org, as the primary employer of First Aid Instructors, is responsible for coordinating continuing unit First Aid Training;

d. Provide effective First Aid and communicate with SMA. This relates to providing immediate treatment and communicating the status of the patient to the SMA; and

e. Stabilizing the patient with the assistance of the SMA and preparing the patient for evacuation to definitive medical care. This relates to providing ongoing treatment to keep the patient stable and preparing the casualty to evacuate to the next level of care.

302.1 Support to the DC Effort

1. Initial Reaction. Everyone is a sailor first, therefore the first reaction to battle damage is the same throughout the ship:

a. Rapid Survey. The first reaction to ship damage is the rapid survey. The CCT members are expected to conduct a rapid survey as soon as possible after damage. If the order to brace for shock was given, then the rapid survey would commence immediately upon receiving the order to unbrace. Rapid surveys conducted throughout the ship and the ensuing reports allow the DCO and Command to get an immediate picture of where there might be fires, floods, structural damage or other incidents of concern. Rapid surveys involve a quick (under 1 minute) set of rounds in predetermined locations and reporting information about any events and initial reaction through the CCT Net for furtherance to Command (or HQ1 directly if possible). If a space is inaccessible (due to smoke, fire, flood or for any other reason), rapid survey is still reported complete with the exception of excluded spaces.

b. Rapid Response. During the rapid survey, CCT members are expected to react to DC incidents like anyone else in the ship. CC members shall conduct rapid response to fires, floods, and other incidents in their respective zones. Similarly, if CCT personnel encounter a DC incident prior to closing up at their CC location, they shall conduct rapid response and only close up when secured. Whether rapid response is successful or not, the incident as well as the result of rapid response shall be reported to the SMA and Command.

c. Boundaries. CCT members are expected to set up initial boundaries in their areas of responsibility to help box in the DC incidents. If the DC incidents are happening more than one zone away from where the CCTs are closed up, CCT I/Cs must assess tactical situation to decide whether or not CCT should provide the boundary. CCTs must be careful not to allocate too many resources to boundaries/sentries if the tactical situation may require patients to be treated. The SMA should expect the CCT I/Cs to report when boundaries rigged by the CCT are set. HQ1 should expect the SMA to report directly to Command on the CCT net when boundaries are set, and then request a relief be sent so the initial boundary personnel can return to the CCT to be available for patient management.

2. DC Picture Awareness. Information flow regarding the DC status and actions taken to deal with DC incidents is a key enabler for the CCT members, especially in the early stages following an attack or incident. CCT members must listen to the XO's SITREPs to ensure that quality control is provided for the internal battle picture and correct any discrepancies immediately through the CC HQ. Maintenance of the DC picture is critical for the safety of the teams and to inform them of safe evacuation locations and routes.

3. Communication. Communication is the key to avoiding confusion in tracking the casualty situation onboard the ship. Casualties need to be quickly identified and their movement up until evacuation off the ship closely monitored. One of the key points is to talk to a conscious casualty and ask them their name, where they came from or ask a rescuer as part of the handover of care from the first "buddy" who is looking after them until a CCT member responds to take over. Contacting the CC HQ as soon as possible after the initiation of care to give their name, explain the status and current and past location of the casualty is vital.

4. Provision of Medical Care. CCT members have an obligation to maintain their skills and provide the best First Aid they are trained to provide to their patients. Provision of quality care provides an essential morale boost in combat scenarios if sailors know they will get the best care available if they need it. If they have an idea of the type of casualty they are responding to, it may guide the equipment they should bring for an initial response.

302.2 Assess the Casualty

1. The first step to properly support the patient is to assess the area that the casualty is in for hazards. The rationale for this assessment is twofold:

   a. It allows you to evaluate any hazards to yourself and the patient; and

   b. Gives you an idea of how far you need to move the casualty to if it is unsafe.

2. The complete steps in the assessment of the casualty are as follows:

   a. Assess the area you are responding to;
   b. Have someone contact CC HQ with name of casualty and where you are located;
   c. Receive a debrief from the first responder to the casualty;
   d. Perform initial First Aid assessment and interventions (MARCHE) to include;
      (1) Massive Bleeding (stop massive bleeds);
      (2) Airway (assess the airway);
      (3) Respiration (check for breathing);
      (4) Circulation (check for pulse);
      (5) Hypothermia (cover patient to prevent heat loss); and
      (6) Everything else (rapid body survey).

   e. Communicate your findings to CC HQ and assess the triage priority to the patient (see 405.1);
   f. Complete Secondary Survey and MIST.
   g. Prepare your patient for movement/evacuation; and

h. Maintain follow up and management of the casualty, reassess periodically and report status changes to the SMA.

302.3 Casualty Evacuation

1. The medical facilities afloat in the RCN are not capable of managing serious casualties for long periods of time or with serious injuries. All casualties will require some form of evacuation or repatriation to obtain the best level of treatment for the injuries/illness sustained.

2. Close coordination with the SMA between ORO, LOG O (if embarked), and Command will be required to coordinate this .

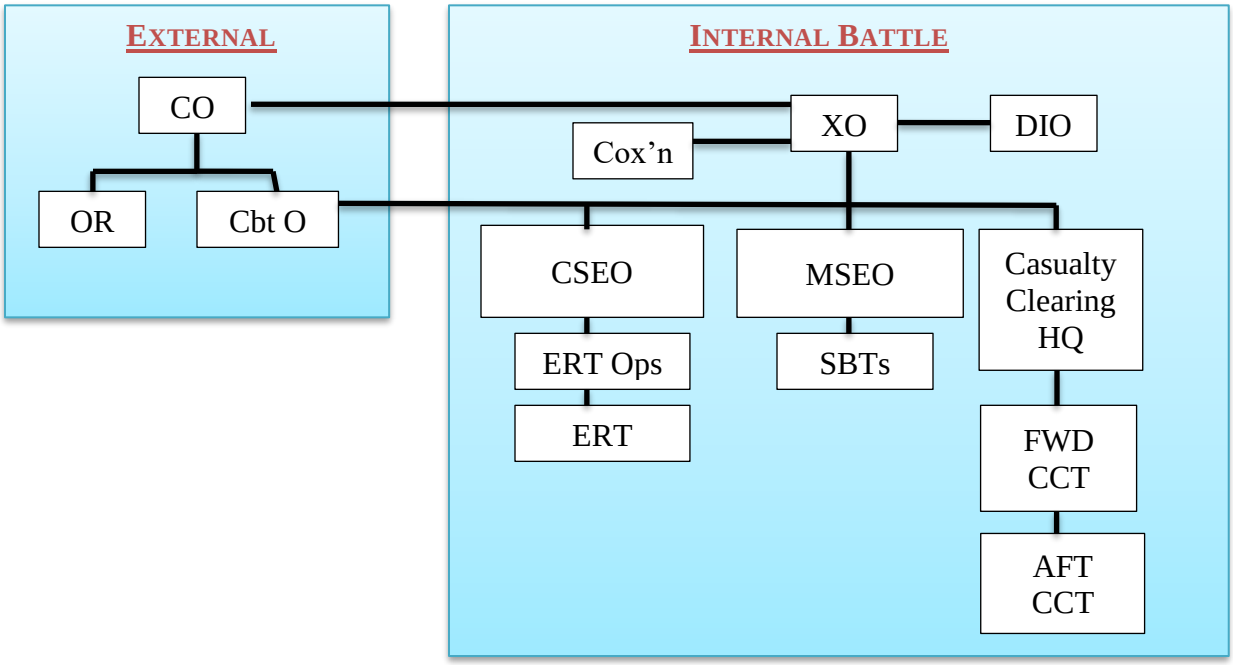

Figure 1 - The Internal Battle Organization C2

303. CCT Organization

1. The CCT Org is typically composed of personnel allocated from the Logistic (Log) and Executive (Exec) departments. To fulfil its duties, the CCT is normally broken down into the following components:

a. CCT Headquarters (CC HQ);

b. Forward CCT (FWD CCT); and

c. After CCT (AFT CCT).

2. The Watch and Station Bill dictate where Log personnel are to respond in different situations. The SMA are responsible to the CO for the medical care onboard, however during occasions when the CC Org is closed up, the CDC acts on the SMA behalf and passes the information through the XO to the CO.

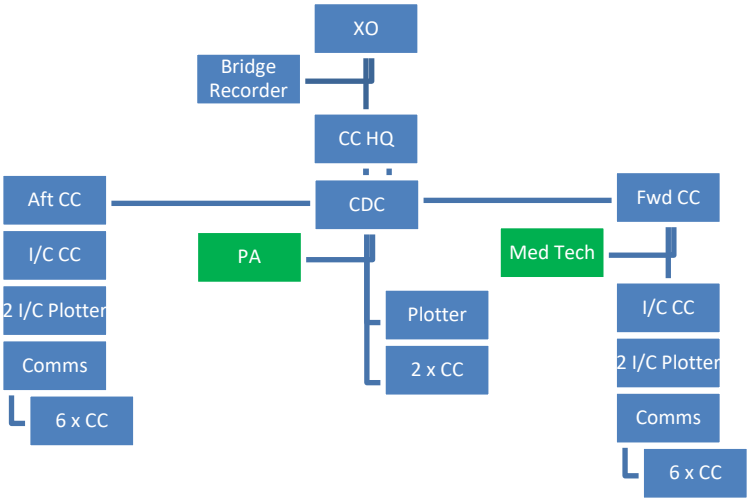

Figure 2 - Typical Close up Locations on Major Warship

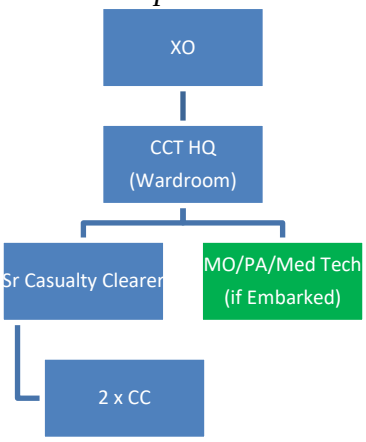

Figure 3 - Typical CCT Org on Patrol Vessel

303.1 Sick Bay/Casualty Clearing Team HQ (CCT HQ)

1. CCT HQ is the control position for the CCT organization and is generally situated in the Sick Bay. CCT HQ is comprised of the personnel and equipment necessary to monitor, record, coordinate, and direct the activities of the CCT outstations. CCT HQ receives direction from the XO through informative pipes on the location of potential casualties. In addition, information is also received on the general DC picture through the informative pipe process and updates to the BDCS in the case of CFP DC scenarios.

2. CCT HQ is normally manned as follows on a CPF:

a. The CCT is under the direct control of the CDC located in CCT HQ. As the CCT Leader, he/she is in communication with Command on the Medical Net, and consults with the SMA when they are available and not treating patients;

b. The SMA, provides guidance to the CDC, when available that directly affects patient care. The PA also ensures that the patients are triaged and treated accurately and command is notified of the casualty situation as the DC picture evolves;

c. Two CCT members (usually an experienced Steward) is in the Sickbay to act as a runner in communication failures, bring unique equipment to the scene of a casualty, act as an escort for the SMA if needed or to provide assistance to the SMA if a casualty is being managed in the Sick Bay treatment facility; and

d. The Plotter monitors the Med Net, maintains the BDCS and provides recommendations to the CDC based on the current DC picture.

3. The CCT HQ primary and secondary locations for each class of ship as well as the related communication setup are included in SSO.

303.2 Forward CCT (Fwd CCT)

1. Depending on the particular equipment layout of the vessel and the available manning, the Fwd CCT is typically mustered in the Wardroom. The Wardroom is also the Alternate Sick Bay location for most classes of ship should the Sick Bay become unuseable.

2. Fwd CCT is manned as follows:

a. The Fwd CCT IC, normally an experienced PO1 from the Log Dept, coordinates and directs the CCT personnel as directed by CCT HQ, providing the forward part of the ship with CC teams as required. The IC needs to maintain situational awareness of the DC situation and casualty situation to make sure their pers are safe as practical;

b. The Fwd CCT 2I/C, normally a less experienced PO2, assists the Fwd CCT I/C in his/her duties while monitoring communications and updating the situation on the BDCS if the ship is equipped. They provide recommendations to the Fwd CCT I/C based on considerations related to the DC incidents. They also must maintain situational awareness and assist the IC in maintaining the DC picture;

c. The Med Tech is another resource available to the SMA to rove the ship and monitor casualty management as directed. The Med Tech must maintain situational awareness and be proactive in reaching out the CCT members out looking after patients throughout the ship. The med tech should initially be looking for the most seriously injured and making sure they are identified as such as soon as possible;

d. Some ships will utilize a separate communicator as well, but with the addition of BDCS and the resulting aggregation of data there, less communication is required over the voice nets; and

e. 6 CC Personnel to make up, three teams, or a suitable ratio of CC teams consistent with the concentration of personnel in the forward end of the ship.

3. The Fwd CCT primary and secondary locations for each class of ships as well as the related communication setup are included in SSO.

303.3 After CCT (Aft CCT)

1. Depending on the particular equipment layout of the vessel and the available manning, the Aft CCT will be mustered in a location in the after part of the ship in an area that provides space to manage casualties and maintain communications with the CCT HQ. On CPFs this is the CSE/MSE Office and CSE Flats area.

2. Aft CCT is manned as follows:

a. The Aft CCT IC, normally an experienced PO1 from the Log Dept, coordinates and directs the CCT personnel as directed by CCT HQ, providing the after part of the ship with CC teams as required. The IC needs to maintain situational awareness of the DC situation and casualty situation to make sure their pers are safe as practical;

b. The Aft CCT 2I/C, normally a less experienced PO2, assists the Aft CCT I/C in his/her duties while monitoring communications and updating the situation on the BDCS if the ship is equipped. They provide recommendations to the Aft CCT I/C based on considerations related to the DC incidents. They also must maintain situational awareness and assist the IC in maintaining the DC picture;

c. Some ships will utilize a separate communicator as well, but with the addition of BDCS and the resulting aggregation of data there, less communication is required over the voice nets; and

d. 6 CC Personnel to make up, three teams, or a suitable ratio of CC teams consistent with the concentration of personnel in the after end of the ship.

3. The Aft CCT primary and secondary locations for each class of ship as well as the related communication setup are included in SSO.

304. CCT Personnel Expectations

304.1 General Expectations

It cannot be emphasized too highly the absolute importance of thorough system, equipment and drill knowledge *amongst the whole of your ship's company.*[5]

1. All CCT personnel are to recognize the critical interface they serve between the personnel morale and operational effectiveness in the RCN. The effective management of personnel casualties directly supports all portions of the ship's operational effect and it is therefore imperative that all CCT personnel understand the added dimensions of the following:

a. Operations. CCT personnel must understand how the ship is generally operating. A ship-wide perspective will support a more effective CCT organization in emergencies and in action;

b. Current mission(s). CCT personnel must comprehend the current mission, how they are supportive of that mission and the specific impact of a casualty to that mission;

c. Their First Aid responsibilities. CCT members must possess the moral courage to maintain their First Aid skills to the highest level possible, and if unsure ask for extra training prior to an incident occurring. They must be able to triage a casualty accurately, maintain a personal awareness of the DC picture around them and how it affects the management of their casualties ;and

d. Current drills related to general DC problems. CCT personnel must be proficient with the drills (e.g., Setting up a boundary, rigging hoses,

[5] *Fighting the Internal Battle – A FOST Best Practice Guide to BDR C2*, Flag Officer Sea Training, Royal Navy, Jun 2007

controlling flooding, performing rapid surveys, etc) related to the areas of the ship they are expected to perform their duties in.

2. Without understanding these dimensions, the CCT will fall short of meeting the full extent of the Command Intent. Further, CCT members at all levels are expected to be able to do the following:

a. Apply the six Survival Rules[6]:

(1) Know Your Ship,
(2) Contain the Damage,
(3) Manage the Consequences,
(4) Recover Capability,
(5) Sustain the Effort, and
(6) Never Give Up;

b. Understand what medical equipment is in each space and the purpose of each component. From the SMA down to the most junior member, all CCT personnel should be able to determine what potential equipment is available to them and how it is properly used;

c. Provide initial medical care and assessment consisting of:

(1) Assess the area you are responding to;
(2) Contact CC HQ to relay initial information about the casualty;
(3) Receive a debrief from the first responder to the casualty;
(4) Perform initial First Aid assessment and interventions (MARCHE);
(5) Communicate your findings and assess the triage priority of the patient;
(6) Complete Secondary Survey and MIST;
(7) Prepare your patient for movement/evacuation; and
(8) Maintain follow up and management of the casualty, reassess periodically and report status changes to the SMA.

d. Have a detailed knowledge of how to transport casualties from all spaces of the ship to the central medical locations and their alternate locations; and

e. Have a thorough knowledge of the ships fitted DC systems, processes, and procedures to use them.

3. All the leadership positions need to be familiar with the requirements of their "CCT job" prior to proceeding to sea. Members of the CCT should also endeavor to learn the responsibilities of the immediate next position in the CoC as well. Remember there are

[6] C-23-045-000/AG-001 *Damage Control (Sea) Vol 1 Shipboard Damage Control*, Change 1, 17 July 2009

always some experienced members of the CC Organization onboard each ship. Use their expertise and knowledge to prepare well in advance of proceeding to sea.

304.2 CDC Expectations

1. The CDC is responsible to the SMA for managing and coordinating all aspects of the internal battle related to casualties. During Action and Emergency Stations, and when the PA is too involved with caring for patients, they will communicate directly with the XO until the PA is free to brief the CO. He/she is to be the ship's expert on the coordinating movement and tracking of casualties in the ship. The CDC shall, as a minimum, be responsible for the following:

a. Develop and maintain the casualty picture. At all times, the CDC must possess a comprehensive knowledge of the condition and location of casualties throughout the ship;

b. They will maintain a situational awareness of the DC situation onboard and move CCT assets to alternate locations as required to ensure patients and CCT are operating in a safe environment;

c. Communicate the casualty picture to the XO and ensure that the situation reports accurately reflect the casualty situation;

d. Collate and push information to the IMDO and Command to communicate the current casualty situation to the CoC ashore;

e. Provide accurate and relevant advice to Command relating to the impact of casualties to the current and future ship's ability to fight the internal and external battle, specifically if certain key positions are no longer being filled;

f. Provide HQ1 with reports on DC incidents in the vicinity of CCT locations; and

g. Be intimately familiar with the casualty picture in BDCS.

304.3 Senior Medical Authority (SMA) Expectations

1. The SMA provides leadership at the scenes of the most complex casualties. The SMA, is the medical subject matter expert of the department. The SMA is responsible for the training of the CC Org and leadership of the CCT during medical emergencies. The SMA needs to work with the Log Chief and Coxn to ensure their Watch and Station bill is properly filled prior to any sailing. They must be prepared to collaborate with the CDC to ensure that medical care is delivered effectively to the entire ship. They are responsible to the CO for this.

2. The SMA is a critical CCT success enabler. He/she has a unique autonomy and serves as a rover with the freedom to do what is required to meet the casualty management needs of the ship. The SMA is the CO's expert support in CCT HQ, providing a slightly detached hands-off auditing function of the CCT picture, as well as an essential link between CCT HQ and the scenes of action where CCT members are conducting first aid to serious casualties. He/she continuously conducts situational assessments and directs execution of plans based on the XO's intent without needing to wait for CCT HQ to pass information via the communications nets.

3. The SMA is responsible to the XO for the following:

a. Ensure the efficient operation of CCT activities in collaboration with the CDC as dictated by the Commander's intent and ship's command priority;

b. Keep the XO fully informed of all situations, especially by reporting through the CDC any status updates relating to ongoing casualty situations, including evacuation implications;

c. Have an intimate understanding of the current tactical situation and the current CCT priorities. The SMA must possess a comprehensive knowledge of RCN management of casualties at sea and must ensure that CCT HQ is made aware of any changes to the priorities of casualties when they occur;

d. Help develop a plan of action to evacuate or hold patients on the ship taking into consideration priorities that are consistent with the external battle requirements;

e. Communicate with shore based medical consultants and facilities receiving patients, to ensure that they are prepared to receive the patients they are sending ashore; and

f. Confirming notification of the RCMS CoC that there have been casualties at sea.

304.4 Fwd and Aft CCT I/C Expectations

1. The PO1 CCT I/Cs are critical to CCT operations as they convey the COs intent (through the XO), maintain the overall casualty picture on the BDCS and directly control CCT outstations.

2. The CCT I/Cs are responsible to the CDC and shall:

a. Muster and report the CCT organization to the CDC when the CCT is closed up in accordance with SSO;

b. Coordinate Rapid Survey reports from their personnel in areas they have assessed;

c. Task and brief CCTs to respond to casualties, as appropriate;

d. Ensure the efficient operation of CCT activities as dictated by the Commander's intent and ship's Command priority;

e. Develop a plan to move and collect casualties in appropriate collection points;

f. Track casualties in their areas of responsibility and validate informative pipes accuracy and correct errors through the CDC;

g. Track and identify priority casualties that need the assessment of the Med Team if embarked;

h. Have an understanding of the current tactical situation and the current CCT priorities;

i. Understand the DC procedures for boundaries, floods, fires etc;

j. Coordinate resupply of consumables and identify shortfalls in equipment with the SMA; and

k. Provide expertise, recommendations and implications to operations and the damage control situation to assist the CDC in determining the requirement to move or relocate resources.

304.5 Fwd and Aft CCT 2 I/C Expectations

1. The 2I/Cs monitor, record, coordinate and direct their respective CCT's personnel as directed by the I/C Fwd and Aft CCT, and provide CCT HQ with the direct communication link.

2. I/Cs and 2I/Cs are the leadership for the CCT outstations (Fwd and Aft CC). They are expected to provide proactive leadership and be aggressive in determining the casualty picture at all times. As well, they must push communication upwards as much as possible so that requests for clarifying information are few. Good leadership is anticipating information requirements in advance and anticipating requirements to support the CC effort throughout the ship. Maintaining the DC picture and providing support to the DC organization through the provision of boundaries, rapid response to DC events and rapid surveys.

3. The CCT I/Cs and 2I/Cs are responsible to the CDC and shall:

a. Be intimately familiar with the layout and situation of the ship in order to develop a plan to collect and manage casualties. This is to include such actions as independently dispatching CC teams to provide initial assessment of their casualty status, communicate this to command through the CDC and recommend an initial triage priority;

b. Provide advice/information related to DC to the CDC for furtherance to HQ1;

c. Provide advice to command through the CDC that impacts the current and future ability for the ship to fight the external battle and the losses that affect the CO's ability to fight the ship, such as critical personnel who have been injured;

d. In conjunction with the IC coordinate resupply of consumables and identify shortfalls in equipment with the SMA; and

e. Be capable of mentally processing several informative points such as verbal reports from multiple sources and information provided by various monitoring systems, to rapidly identify and correlate potential issues to casualty clearing teams' success.

304.6 DC Plotter/Communicator Expectations

1. The DC Plotters maintain the BDC and the POU laptops and provide recommendations to the CDC and Command on the Ship casualty picture. They also provide a clear picture to their team with respect to fires, floods, electrical and structural problems. The DC Plotter must be able to effectively communicate using the BDCS and synthesize the information they are receiving through the medical net, radios and XO pipes, in real time in the system.

2. In the event of a failure of the BDCS/POU laptops, the DC plotter must be ready to switch plotting to a manual stateboard, thereby ensuring the continuous presentation of DC information for the Team.

304.7 Casualty Clearer Team Member Expectations

1. The team member is expected to maintain their First Aid Skills and strive to continually improve and add to their skills. Personnel will be paired with another member of the org to perform their function.

2. Casualty Clearing Team Members are responsible for:

a. Providing accurate information, and prompt reports as information becomes available to the CCT I/C;

b. Conducting accurate triage and assigning the proper priority for medical assessment;
c. Providing medical care within their scope/SOPs and recognize when they need assistance;
d. Assisting the Medical Team (when embarked) providing more advanced care;
e. Maintaining, observing and monitoring stabilized patients before they are transferred off the vessel; and
f. Maintaining the Fist Aid equipment onboard the vessel (in conjunction with the Medical Team if embarked).

3. It is expected that one member will focus on treating the casualty and the other will continually reassess the local situation and maintain communications on the Med Net.

CHAPTER 4 – CCT PROCEDURES AND BEST PRACTICES

401. General

1. This chapter covers procedures and best practices pertaining to the CCT, which are effective in fulfilling the duties of CC Org within the Internal Battle Organization. These form the basic standard of how CCTs operate within the RCN.

402. Closing Up

1. For the CC Org, being closed up is defined as when all CC locations have established the primary form of communications and they have one team with equipment ready to respond to a casualty.

2. The actual manning of the CC Org is found in Class specific SSOs Vol 2.

3. Fully Establish Communications. All out stations must fully test all primary and secondary communications with the CC HQ. This includes Radio, SHINCOM, and BDCS Chat where available.

403. Internal Battle Organization Battle Rhythm

1. To ensure the CC can communicate effectively and efficiently it is important that all out stations have an expectation of a typical battle rhythm on the Med Net.

2. Immediate actions. Once a casualty or damage occurs to the ship, CC out stations will immediately conduct the following:

   a. Rapid Survey – To determine extent of damage and determine any hazards, and search for any casualties in the immediate vicinity of the CC Teams; and

   b. Rounds/Actively search – Understanding the manning of the ship, the CC Org should actively search for casualties in areas identified as damaged. When casualties are piped on the overhead broadcast they should react promptly to the scene and report back to their respective Fwd or Aft CC outstation on what they have found and who the casualties are.

4. Priorities. Once a CC Team has fully assessed a casualty, they are to assign a priority to the patient. It is imperative that it be clear that the priority assigned by the CC Team is the priority for the SMA to assign either the Med Tech or the SMA to personally see the patient and provide intervention. This initial priority is not the evacuation priority. The evacuation priority will be determined by the SMA and transmitted to command when it becomes clear. More clarification on priority assignment will be found at article 405.1.

5. CC outstations must report back to CC HQ on either BDCS or by SHINCOM at a minimum on the following occasions:

a. When a rapid survey had been completed;

b. When they dispatch a CC team;

c. When a Casualty is found (with name, rank and where they were injured);

d. When the CC teams discover damage that has not been reported on the overhead broadcast/XO update;

e. When and where a casualty is moved;

f. When the CCT Has all the information to assign the priority and MIST for the patient; and

g. When the SMA has all the information to assign an evacuation priority for the casualty.

404. Communications

1. Clear, concise and accurate communications within CC Org is critical to the effectiveness of Casualty Clearing activities. It is essential that reports at all levels follow expected norms to ensure that all parties involved have full situational awareness from the CC Team on the scene to the Commanding Officer trying to evacuate his crew.

2. Confusion with CC Org is normally created through the misinterpretation of common terms, and members of the CC Team not being familiar with the ship; therefore it is vital that all members within the CC Org have a common understanding of terms used to describe casualty injuries. A report that is not communicated properly can result in confusion, wasted time and effort, and even a patient not being seen and managed appropriately.

2. Further, communication problems can be exacerbated by the sheer volume of chatter over the net being used. Therefore it is essential that the CC Org be able to synthesize the incoming reports and provide an overall assessment that is clear and concise. In order for this to be achieved, strict NET control must be exercised by the I/C of Fwd and Aft CC. CC Teams must prepare and practice sending clear and concise reports consisting of:

a. Name of casualty;
b. Location casualty was initially discovered and now their present location;
c. MIST; and
d. Priority for medical care.

404.1 Medical Priorities and Triage

1. Whenever you have more than one casualty, the casualties need to be prioritized for initial medical management on the ship. This initial triage priority is assigned by the CC team who respond to the casualty. The assignment of the priority assists the onboard medical team sort who needs to be assessed and managed first to provide the best outcome in a mass casualty scenario. As a general rule mass casualties exist when the severity of the patients outstrip the medical assets available, however in practice mass casualties usually exist in the RCN whenever you have more than one casualty or more than one per MO/PA.

2. The SMART Triage Sieve is a tool that will be used to identify the priority for medical care when Casualty Clearers or First Responders respond to the scene of mass casualties in the RCN. The SMART Triage sieve is used with the SMART Triage cards currently used to record individual patient information in the "field" during mass casualty events for the CF H Svc and as such will also be used by the RCN at sea. This is also IAW STANAG 2122 Edition 3 Appendix 1 to Annex 1. **This triage sieve shall only be used for adults**.

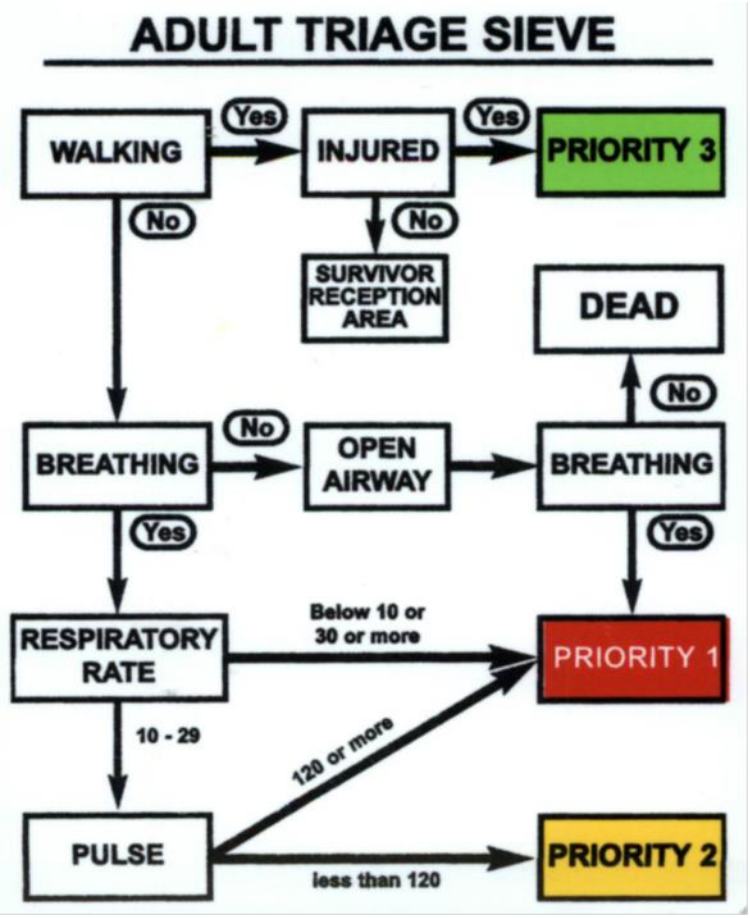

Figure 4 - SMART Triage Sieve

3. The NSN for SMART Card Resupply is 6515-CF-002-1000 and the Sieve is printed on the outside of the package of 10 cards.

4. All casualties must be triaged by the on-scene teams as above to one of the following groups:

a. Priority 1- (casualties that are breathing too fast (over 30) or too slow (less than 10) and have a fast pulse) ;
b. Priority 2 – (casualties with a respiratory rate between 10-29 and a pulse less than 120);
c. Priority 3 - (walking and injured); or
d. Priority 4 - (Dead).

5. **The SMA or Sr Casualty Clearer (in the absence of an integral Medical Team), *or Command if the situation requires*, will then triage the groups to the following evacuation groups once the casualties are stable and the mass casualty situation itself has stabilized.**

6. The following is the list of Naval evacuation triage priorities to be assigned within the RCN (IAW the NATO Handbook on Maritime Medicine (AMedP-11), AMedP – 1.10 Medical Aspects in the Management of a Major Incident/Mass Casualty Situation) by the SMA. The use of the NATO "T" prefix will allow all to confirm that this is in fact an evacuation priority:

a. **Immediate Treatment (Group T1)**

To include those requiring emergency lifesaving surgery. These procedures should not be time consuming and should concern only patients with high chances for survival;

b. **Delayed Treatment (Group T2)**

To include those badly in need of time-consuming major surgery, but whose general condition permits delay in surgical treatment without unduly endangering life;

c. **Minimal Treatment (Group T3)**

To include those with relatively minor injuries who can effectively care for themselves or who can be helped by untrained personnel; and

d. **Expectant Treatment (Group T4)**

This group comprises patients who have received serious and often multiple injuries, and whose treatment would be time-consuming and complicated with a low chance of survival. If fully treated, they make heavy demands on

medical manpower and supplies. Until the mass casualty situation is under control, they will receive appropriate supportive treatment only.

6. Additional Triage categories are used for recompression during SUBSMASH and are included below (IAW ATP – 57 NATO Submarine Search and Rescue Manual):

a. **C1**.

Patients in this category have symptoms of serious or life-threatening DCI and require immediate treatment in the on board recompression chamber. To delay their treatment would entail a significant increased risk of death or subsequent permanent injury;

b. **C2**.

Patients in this category have minor and non-life-threatening DCI symptoms at the time they are assessed. If the scenario involves rescuees from a DISSUB saturated at a DISSUB pressure above 1.7 bar, all rescuees have a decompression obligation and are at some risk of DCI. These rescuees by definition fall into the C2 category and require close monitoring for the development of DCI symptoms. If sufficient oxygen is available, they should receive prophylactic surface high flow mask oxygen treatment until they can be recompressed and treated on a saturation decompression table. *Note: Escapers from any DISSUB or rescuees from a pressurised DISSUB (> 1.5 bar) with no obvious symptoms should be considered C2 and should be held in the tertiary area for medical observation due to the risk of late development of symptoms of DCI*; and

c. **C0**.

Patients in this category have no current indication that recompression is required.

7. Dealing with Mass Casualty incidents requires significant resources, and in cases where there are conflicting demands and resources are scarce (for example, with simultaneous DC problems and Casualties), the CDC will address the requirements with command to supplement the CC Org in order to keep up with the current command priorities. Once DC issues have been satisfied, there should be significant resources available from the remainder of the ships company.

404.2 Communication Plans

1. The CC Org must have communication plans that consider the potential of battle damage affecting the primary communication systems. It is recommended that CC use the acronym PACE as a means to build a robust communication plan. An example of PACE for the Med Net on a CPF is as follows:

a. **P**rimary: MMN 56;

b. **A**lternate: Portable Radio on appropriate channel as promulgated in SSOs;

c. **C**ontingency: BDCS Online Chat; and

f. **E**mergency: Runner.

2. It is important that all members of the CC Org understand not only their primary reporting procedure, but also their alternate, contingency and emergency mode. These alternate communication modes should be routinely practiced to ensure they are understood and do not conflict with other section's communication plans.

404.3 Voice Procedures

1. The CC Teams reports on the Medical Net should be concise and brief. The typical reports that a CC Team makes on the Med Net are so common that CC Teams should practice them before they are required in action. There is no requirement to treat the Med Net like it is a formal ship-to-ship radio, plain language can be used when discussing complicated decisions or implications. However, using common reporting procedures does reduce reporting times on the net and increases understanding by all. Finding the most efficient way to communicate things simply is a challenging task for CC Teams that requires practice. Typically, CC Teams in training require multiple problem scenarios to increase their abilities in this area. Whenever possible, training scenarios that include concise reporting of patient priority and MIST should be included in ships training plans.

2. Much the same as the CC Teams, the CDC and I/Cs fo Fwd and Aft CC must practice the same net discipline. It is the responsibility of the CDC and I/Cs fo Fwd and Aft CC to manage the Med Net and ensure that only pertinent information is being passed and it is reported in a clear and timely fashion. Most CC Orgs use an CC Comm number to pass information for the ERT IC. It is important that these members are clear and concise when using the net.

3. The following are some common reports:

a. "Command, Sick Bay. Casualty Clearing closed up."

b. "Sick Bay, Aft CC. Aft CC Closed up."

c. "Sick Bay, Fwd CC. MS Blood, Moved to Fwd CC."

d. "Fwd CC, Team 2. LS Bloggins removed from steering gear compartment and moved to 17 Mess." and

e. "Fwd CC, Team 2. LS Bloggins Priority 1, MIST, M - Crush injury, I – Fractured ribs, S – Pulse 130 , Respirations 20, T – Splinted, Oxygen. "

404.4 BDCS as a communication tool

1. BDCS, when installed, has the capability of augmenting communications during a communications breakdown. Within the system there is a "chat" function which should be monitored at all times.

404.5 Recording Casualty Information

1. Casualty information must be maintained and accurately recorded during all Casualty incidents. In Mass Casualty incidents this is even more important as the SMA must assemble all information into the individual casualties' medical record and transmit this same information to the receiving facility promptly and accurately.

2. Casualty information is to be recorded on official medical records and BDCS if it is available and not into "notebooks" or on scraps of paper. The temporary use of official notes and logs to transcribe into the official documentation is permissible as long as the notes are legible, stored appropriately, and maintained as official records in the case of an investigation. Personal notes are not permitted.

4. Individual patient records are to be maintained on the official CF Patient record documents provided by CF H Svcs. Local unauthorized solutions are not to be used.

5. The current record is the SMART Triage Card NSN 6515-CF-002-1000 (Copy attached at Annex A). For training purposes, it is acceptable to scan a card and use a colour printed copy of the card to document exercise casualties.

404.6 Communicators

1. In order to provide consistency in controlling the Med Net, personnel should be designated and practice communicating on the Net. On ships with BDCS the communicator should also be updating and plotting BDCS once proficient in both pieces of equipment.

2. On ships where the communication is with SHINCOM, the Communicator should use a headset and place the SHINCOM on "Speaker" as well, so the I/C and Teams can monitor the situation and comms. Also, the appropriate DC net should also be accessed on the terminal to report DC issues to the DC Org directly. At a minimum, the CC HQ shall have a method to report DC problems to the DC Org.

404.7 Command Conferences

1. At some point in time the SMA needs to confer with command to discuss the overall casualty situation in a casualty incident. Preferably the SMA should discuss the situation directly with the CO as soon as the situation stabilizes. The XO and Coxn should be engaged early to facilitate the process of evacuation or onboard manning requirements to maintain mass casualties onboard the ship. The Ops O also needs to be appraised of the evacuation priorities as soon as they are available so they can investigate the various option available to the CO for management of the situation.

404.8 Situation, Impact, Options, Recommendations (SIOR)

1. When reporting the situation to the CO, the SMA should use the SIOR format. If there is not sufficient information to provide adequate options or recommendations, only the situation and impact need to be reported. The impact should include actual and potential impact.

2. The SMA needs to ensure that they explain the casualty situation and the impact on the patient prognosis, while remembering that the CO needs to make the final decision on when and how casualties will be evacuated back to Canada if required. It is sometimes useful to ask the ORO/OPS O to accompany you with the Cox'n and XO to brief the CO.

404.9 MIST (Mechanism of Injury, Injuries, Symptoms, Treatment)

1. When reporting casualty information to the appropriate CC Team the following format will be used:

a. Patient Name (**and** casualty number if known);
b. Location of the casualty and where they came from;
c. M – Mechanism of injury (Fall, blast injury, crush, etc…);
d. I – Injuries sustained (Fractured arm, head injury, Burns, etc…);
e. S – Symptoms (at least level of consciousness, pulse, and respirations. Ensuring that Pulse and Resp is always recorded into the system will allow the I/Cs and SMA to confirm the correct triage has been assigned, and spot any trends for worsening that may occur);
f. T – Treatment given so far (Oxygen, splint, tourniquet, etc…); and
g. Patient Priority (Calculated by the Sieve).

405. Plotting Casualties

1. For ships equipped with IPMS/BDCS the BDCS System is the primary plotting method for plotting, tracking, and generating updates to the casualty picture in a mass casualty incident. When BDCS fails, it will maintain the picture on that specific POU at the time of failure so the plot will have to be maintained manually. The same as if the POU is relocated.

2. For ships without the BDCS, the manual plotting board method will be used, and will also be a secondary backup to the BDCS in times of hard failure.

3. Maintaining the plot is the primary tool for the CC Org and Command to do the following:

a. track and collate casualties;

b. identify priorities for managing and locating casualties;

c. assess impact of casualties to the mission;

d. Understanding how DC events throughout the ship will impact CC Org objectives;

e. identify manning shortages; and

f. assess reduced capabilities due to casualties.

405.1 Manual Plotting Board Content

1. All manual Casualty Plotting boards will have the minimum capabilities as follows:

a. Ships Plan. A diagram with the ships compartment layout and organisation, including the locations of CC equipment and assets and areas that CC Org is responsible for during rapid surveys;

b. Close up Reports. Blocks for close up reports from CC HQ, Fwd CC, and Aft CC as appropriate to the class of ship;

c. List of casualties being tracked. A table that includes the casualty number, name and rank of the casualty, location of injury, present location, MIST, and the two priorities (One for medical care, and the other for evacuation), and who is assigned the casualty (Fwd or Aft CC). A log book can be used with the above information if there is no room on the board, however it needs to be legible and large enough to be read "over the shoulder" if other pers have a need to the information;

d. Team Composition list. A block set aside to manage CC Team accountability and identifying where they are at all times; and

e. Initial Actions checklists. For Rapid Survey, Initial assessment of casualties.

2. Ships are highly encouraged to have additional materials and memory aids available to be used by the whole team. Simple aids for plotting casualties should be on the board.

3. The Manual Plotting board can take the form of a laminated sheet and is not required to be in the traditional board format, as long as it is large enough to be read by a group of people around it and it is able to be annotated with information legibly.

405.2 Plotting Board Management

1. Managing the plotting system onboard consists of the following:

   a. Pre-Sail maintenance. Both the manual plotting board and BDCS need to be reviewed at the beginning of any sail to:

      (1) update the teams and allocation. In BDCS the teams for each CC station need to be defined. On the manual board the teams should be identified in a list; and

      (2) ensure that the plots are ready for emergency use. The lists of casualties have been cleared and the DC state zeroed.

   b. Post-Incident Maintenance. The Board, Casualty logs, and BDCS information from each section of the CC Org are to remain intact and under control of the CC HQ, until evidence gathering is complete and then can be reset and zeroed. Spare manual boards are to be available in the case of ships without BDCS so evidence gathering does not impede the use of the board for subsequent issues.

405.3 Casualty Notification and Plotting Convention

1. There are three ways that casualties are typically identified during an incident:

   a. From an informative "Casualty Pipe" or overhead broadcast;

   b. A casualty "just shows up" at the CC location during an emergency;

   c. Another shipboard organization plots a casualty in BDCS (Should only occur in a breakdown in comms); and

   d. Bystanders shout "casualty at *location*" and word is passed verbally throughout the ship.

2. All casualties will be matched to their name, rank and injuries as soon as possible in order to maintain situational awareness and accountability of the crew. For example, not entering/cross-referencing the name of a crewmember who is a casualty will result in section leadership not being aware of where all their missing people are, and could result

in a missing persons search or verification muster being initiated in the resultant confusion. Knowing the name also can help clarify situations where the casualty is moved many times throughout the ship before the CC Org starts managing them. A casualty properly identified in BDCS also allows command to account for pers with special responsibilities/positions.

3. **Manual Plotting**. When a casualty is identified:

a. all stations will enter the information in the manual logs as they receive it to provide redundancy, and to build the manual picture;

b. the casualties will then be plotted and matched with the team responding, by the outstation who will then automatically dispatch teams according to their pre-designated area of responsibility without input from CC HQ;

c. when the team locates the casualty they will inform the CCT that they have arrived and give the name of the casualty to be recorded in the log, and will follow up with the rest of the required information when it becomes available;

d. the Plotter will then enter the information onto the plotting board as it comes in;

e. as the casualty is moved through the ship, they will be tracked along with the team responding to each casualty; and

f. supplemental information about each casualty will also be manually tracked.

2. For example; on a CPF, the Fwd and Aft CCT division of responsibility is at Frame 34. Fwd CC is responsible for the forward end of the ship and Aft CC is responsible for the after part of the ship. A casualty is identified in the OPs room. All stations log the casualty, and the Fwd CCT I/C dispatches and logs Team One to the first casualty. As updates come in they are added to the plot.

3. **BDCS Plotting**. Ships with BDCS, are to ensure that on slipping the jetty, that all the CC Org Team members are initially loaded into the system for manning accountability.

4. On close up, all the teams that are available are to be plotted in the system as ready for dispatch in the location of the CCT Fwd or Aft. This step done in advance also makes it very easy to dispatch them to the scene of a casualty in the BDCS, you can then just drag them around the ships plan.

5. As casualties are initially identified:

a. The CDC will cause the CC HQ BDCS POU to display the initial location and number of casualties as appropriate. (Figures 5 And 6) All stations will have

the capability to do all CC HQ plotting functions however; only one station will initially identify the location of a casualty in order to not create duplicate casualties in the system. This will typically be the CC HQ unless there is a network failure in which case the CDC will authorise the appropriate CCT to input the initial casualty over the net. Ensure all alternate POUs can plot casualties and update casualty information;

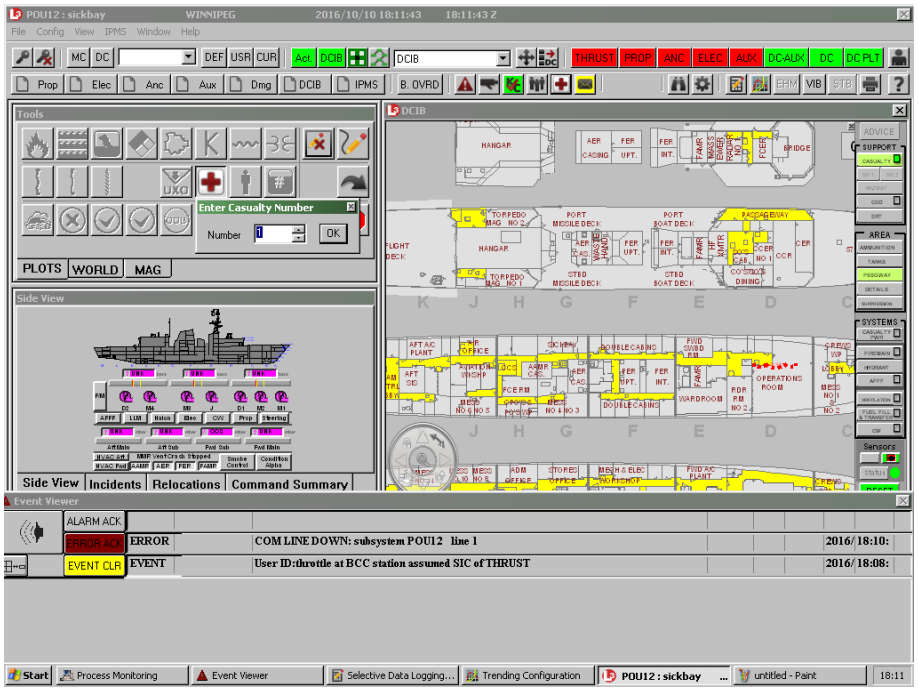

Figure 5 - Casualty Plot

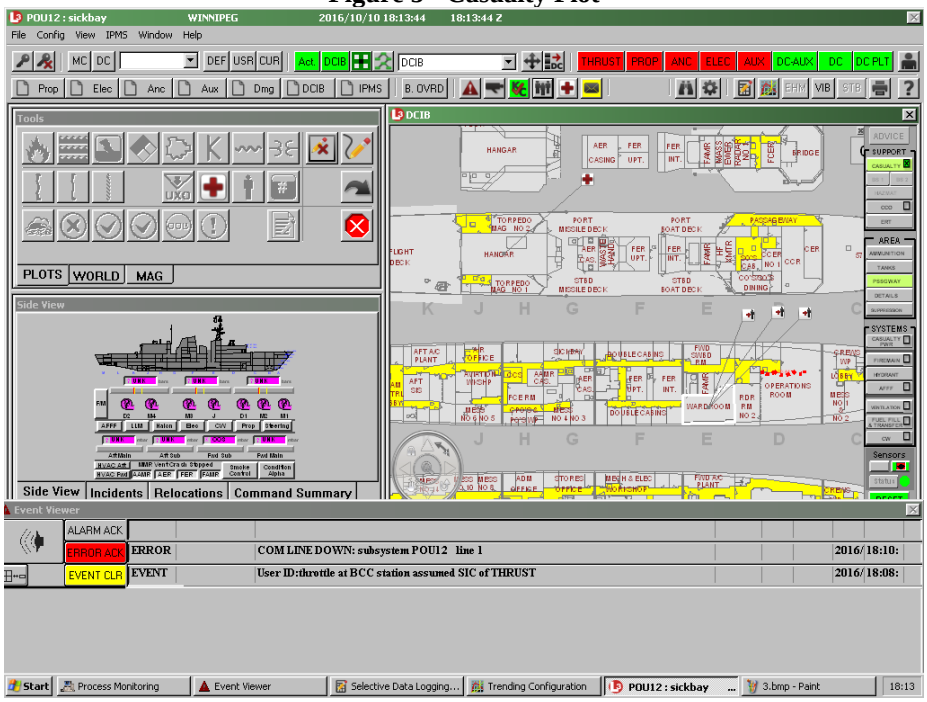

Figure 6 - Casualty Location

b. Once the casualty is initially identified in the BCDS, the appropriate CCT is then responsible for plotting everything related to that casualty and all remaining updates to information;

c. The Fwd or Aft CCT will dispatch and log a team in the system as responding to that location without the need for prompting by CC HQ (Figures 7 and 8) . **There is no need to delay the team response while the entry is made into the system**;

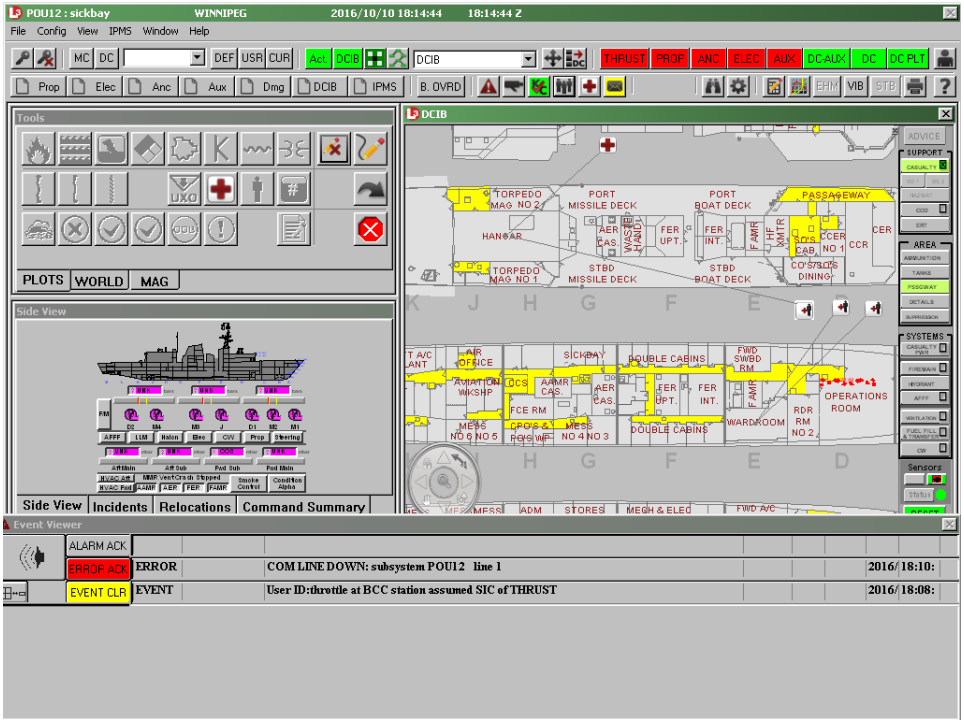

Figure 7 - Match CCT to Casualty

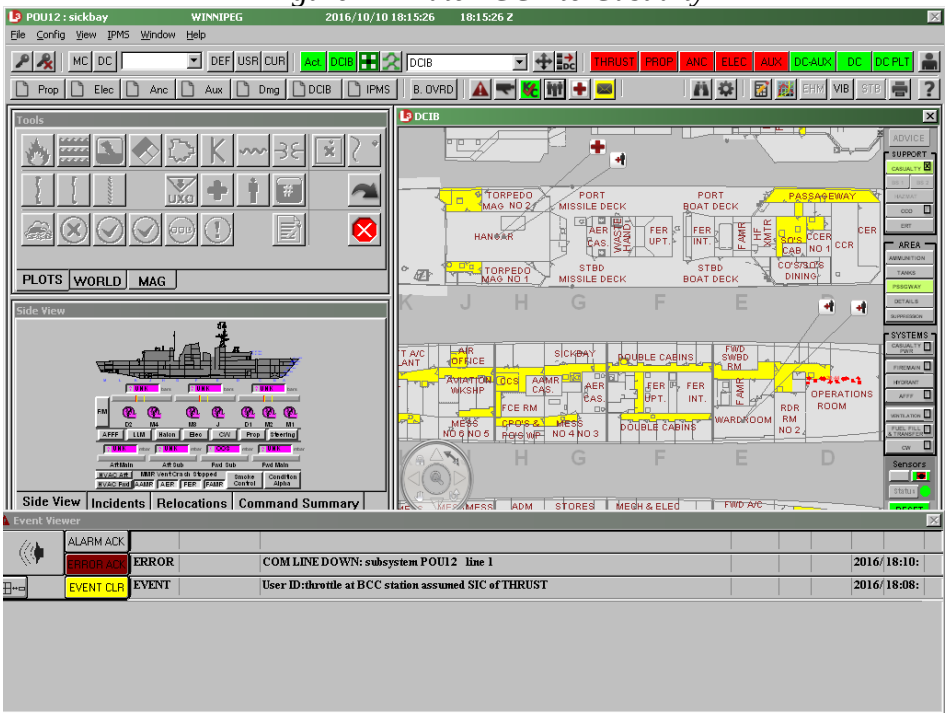

Figure 8 - Tidy up Icons and keep CCT and CAS Icon Together

d. Once the team locates the casualty, the team will communicate that they have found the casualty and give the name and present location as well as where the casualty came from/where they were injured. The appropriate CC Plotter will

then update BDCS with the basic information (Figures 9 and 10);

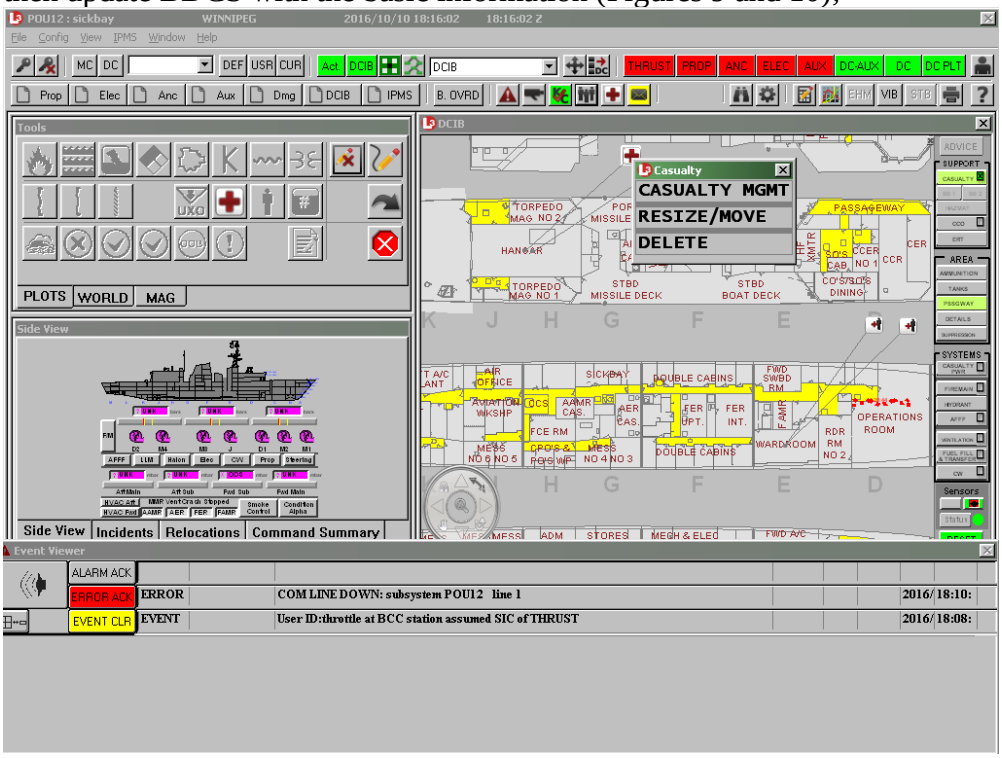

Figure 9 - Update Basic CAS information (Part 1)

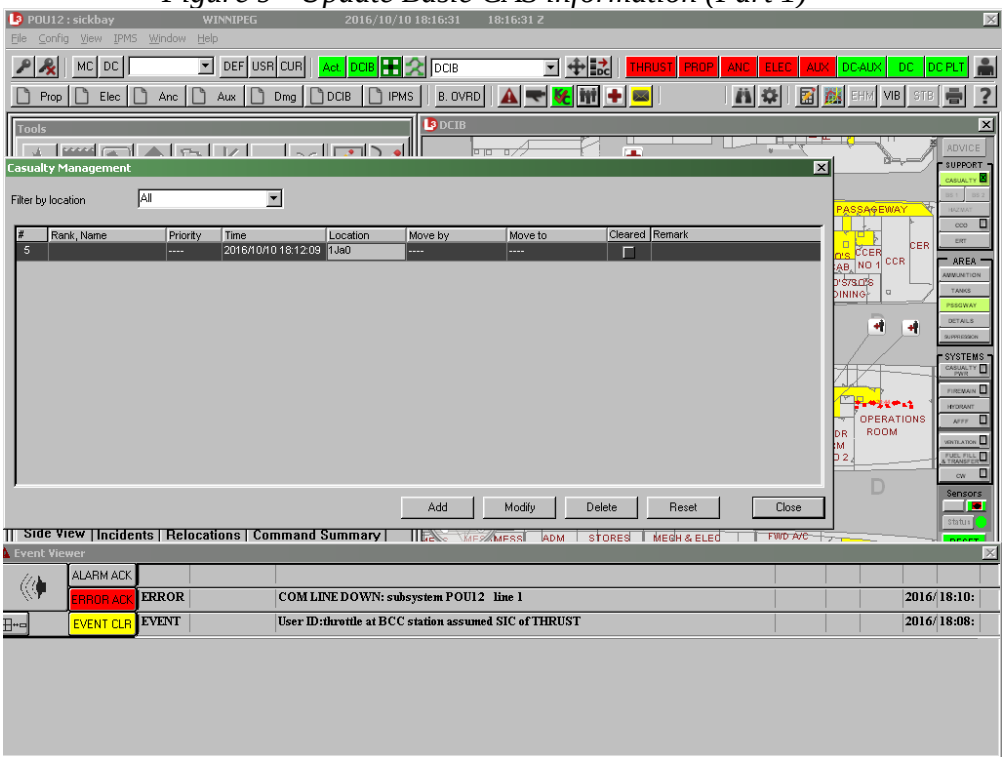

Figure 10 -Update Basic CAS information (Part 2)

e. The team will continue to manage the patient and gather information to supply the patient priority and MIST and then will contact their CC location with that information when it becomes available; and

f. The BDCS Plotter in forward or aft as appropriate, will then enter the information into the casualty record/log in BDCS (Figures 11 and 12).

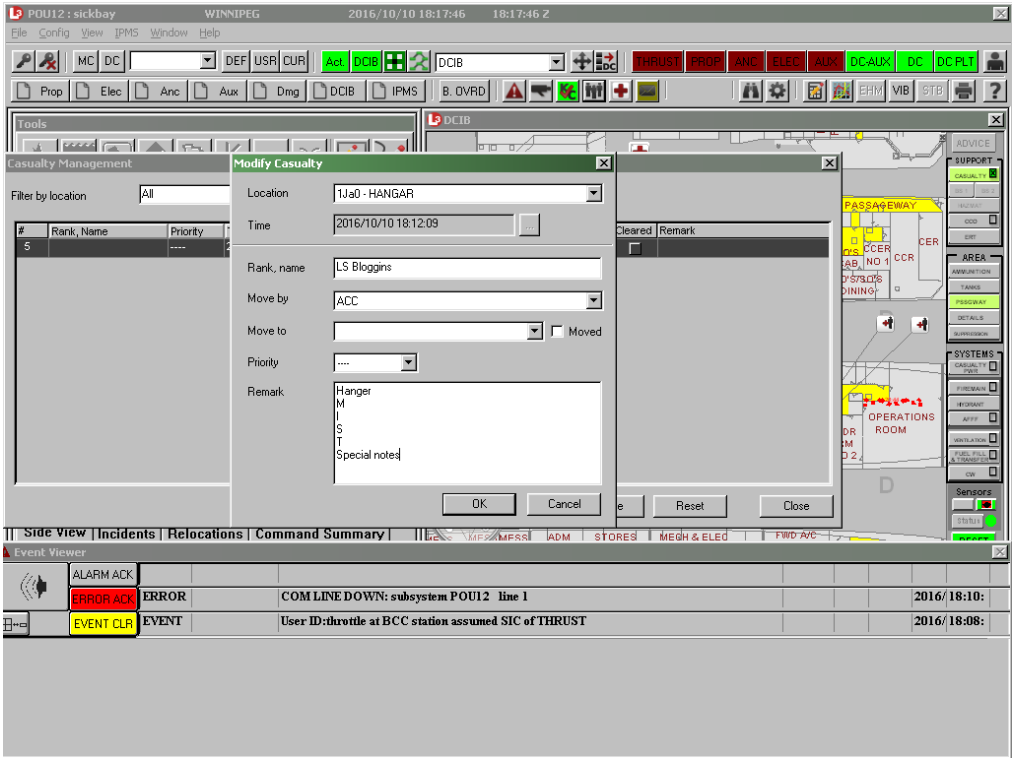

Figure 11 - Complete Casualty Information (Part 1)

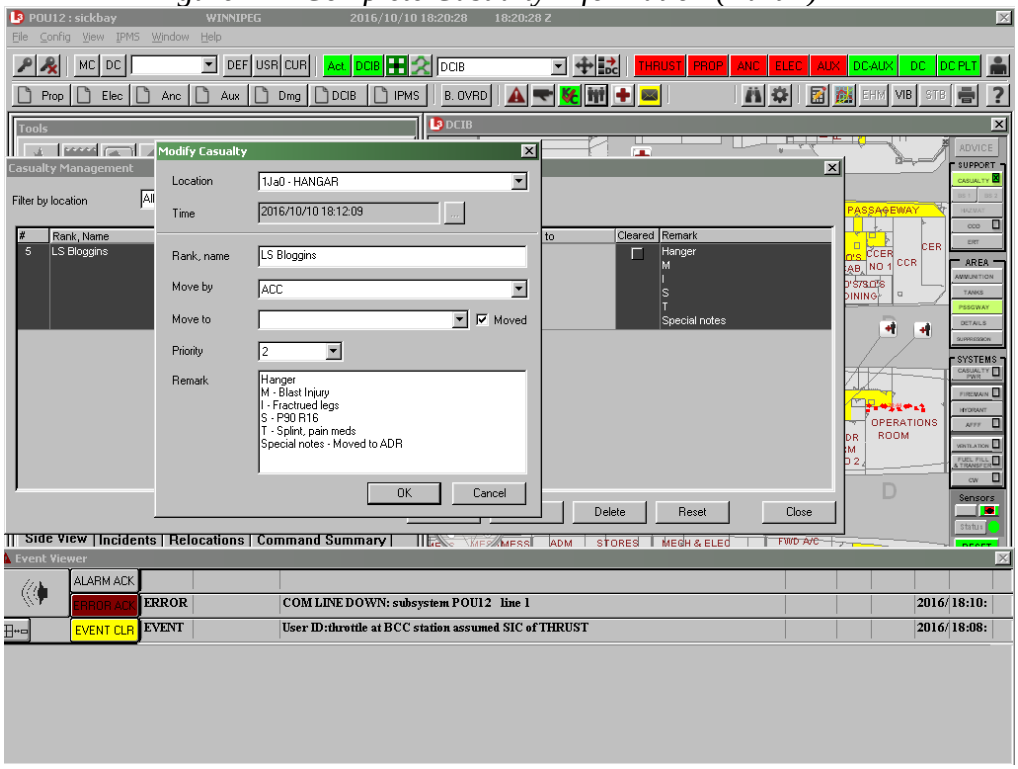

Figure 12 - Complete the Casualty Information (Part 2)

6. All casualties will have the following recorded into BDCS and updated with any changes to status:

a. Name;
b. Rank;
c. Casualty Number;
d. Priority;
e. MIST(symptoms must be complete with at least Pulse and Resp rate); and
f. Present location.

406. Actions when Casualties are identified

1. The first reaction to the realization that the ship has sustained damage is rapid survey. All CC outstations shall conduct a rapid survey when ordered to do so. If the order to brace for shock was given, then rapid survey would commence immediately upon receiving the order to unbrace. Rapid surveys conducted throughout the ship and the reports that come after those surveys allow the DCO and Command to get an immediate picture of where there might be fires, floods or other incidents of concern once battle damage has been sustained. In the context of the CC Org, rapid surveys involve a quick (under 1 minute) set of rounds to find any DC incidents within the designated area determined by the senior hull tech, conducting an initial rapid response to any incidents, and passing information about the event and initial reaction through the Med Net for forwarding to HQ1 on the Command Net.

2. When the ship has sustained damage it is important to expect casualties and actively confirm if any exist. If no casualties are "piped" on the overhead broadcast, then it is not unreasonable for the I/C of the area CC Teams to send out a team to investigate the area of damage to confirm nobody has been injured, especially if the area is known to normally have personnel in the immediate vicinity.

3. A team of Casualty Clearers is two personnel with their first aid bags. They will respond to the scene of a suspected casualty and report back to their I/C once they have made initial contact.

4. If no casualty is found initially, the team will search all the adjacent spaces and secondary zones and investigate if the person was moved by other personnel in the area. If no casualty is in the area after that, they will report back to the I/C for further assignment.

407. Casualty Rounds

1. Once an event has occurred, as soon as a few casualties have been located, the SMA (and Med Tech if embarked) are expected to go around the ship and "set eyes" on each patient once the DC incident stabilizes and the casualty picture stabilizes.

2. They are to maintain communication with the CDC while they are doing this.

3. The primary purpose of having them conduct rounds is for them to gather information about the casualty picture and to initiate treatments and procedures that the Casualty Clearers are unable or untrained to perform. They are also to reassess the status each casualty to determine the evacuation priority.

408. Personnel Accountability

1. Care for the health and safety of personnel is one of the most import things CC I/Cs are responsible for. CCT I/Cs are to ensure the following:

a. ensure personnel are alert enough to perform their given tasks;

b. personnel hydrate;

c. maintain a sustainable work/rest cycle;

d. Log their location and movements throughout the vessel; and

e. ensure personnel are properly briefed on what their objective is, routes if they need to navigate danger areas such as fires/hazards in the area and the acceptable amount of risk they may expose themselves to in order to complete their objective.

2. It is the responsibility of the CDC and CCT I/Cs to mitigate risk to personnel whenever possible.

409. Casualty Transportation

1. In general, common sense is to prevail in the movement of casualties throughout vessels. There is no prescribed number of personnel that are required to move the casualty as long as common sense and general safety concepts are used. The Universality of Service Requirements[7] require that all CAF personnel:

> "Must be able to evacuate a casualty 750 m across country by carrying one end of a stretcher bearing an 80-kg load.[8]"
>
> "Must be able to evacuate a casualty, during a fire on board ship, by carrying one end of a Stoke's Litter bearing an 80-kg load up and down a flight of stairs"

That being said, the minimum number of personnel to move a stretcher is two personnel. However, in certain sea states and for certain movements more personnel may be required depending on actual conditions at the time of transport. Again, common sense needs to be the guide.

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[7] DAOD 5023-0, Universality of Service, and DAOD 5023-1 Minimum Operational Standards Related to Universality of Service

[8] A-MD-154-000/FP-00 Medical Standards for the CF

2. The following conventions apply unless the on scene supervisor with due consideration believes that modification is required for the sake of safety:

a. Prior to evacuation taking place, the Sr Casualty Clearer present will identify themselves and take responsibility for the planning and safety of the movement;

b. The Sr Casualty Clearer will ensure that all lines attached to the patient and stretcher present are secure, all equipment attached to the patient or stretcher are attended or securely fastened, and that personnel are assigned to track and monitor the vital signs of the patient at regular intervals. At a bare minimum, they are responsible to ensure that before any vertical movement between decks, vital signs will be assessed and all straps checked for security;

c. Casualties will be transported feet first whenever possible, when travelling horizontally;

d. Casualties will be transported head first when being transported vertically, unless space allows for them to be transported horizontally (ie: over the side of the ship in the Billy Pugh, or in a Boat);

e. If you need to stand on a ladder to protect the casualty from rotation(and a subsequent smashed face!) and hand them up one deck, and it is safe to do so, carry on;

f. A CC Team Member only requires one hand on the stretcher so that one hand is available to them to steady themselves in rough weather;

g. A handling rope is only required on the head end of a stretcher when transporting vertically, as long as the distance is short enough that the bottom of the stretcher can be maintained under positive control by the pers on the lower deck. If the movement is short enough and positive control is possible, it may even not require a rope at all; and

h. Sometimes two people transferring a stretcher is safer (and often faster) than four, be aware of constricted spaces that fewer people may work better and adjust accordingly for the situation.

3. **One absolute safety concern** is that no patient will be transported over water, in a boat or aircraft, or adjacent to guard rails where there is any possibility at all that it may fall to the water, without ensuring sufficient buoyancy and stabilization is attached that will cause the stretcher to remain in a head up and face up position if it falls accidentally into the water[9].

[9] NATO STANAG 1412 and AMedP-1.4 Transfer Litters Ship to Ship or Ship to Air

409.1 Packaging of Casualties

1. Packing of casualties in stretchers will conform to the manufacturer's instructions in as much as the safety and security of the patient is concerned. In addition, the comfort of the patient is not to be overlooked. Pressure injuries can start to occur very early on in an immobilized patient and prevention is the key to avoid long term issues. Extra padding is always required when the patient will be in the stretcher for some time. At an absolute minimum a folded soft blanket will always be against the patient, and patients will not be placed directly in contact with the stretcher for more than 45[10] minutes without appropriate padding.

2. When at all possible, the principle of minimum handling will apply[11]. It is preferable at the earliest possibility to place a casualty on a clean (dry, strong, lint free) sheet first, and use that to lift and move the patient, or use a transfer device such as a "scoop style" stretcher. This is especially important for burn patients, to avoid further damaging fragile tissues. The patient should be exposed and all items such as knives keys etc., which could cause additional pressure points inside the stretcher, shall be removed before binding the casualty up in the stretcher. Keeping a suitable clean sheet in/with all the stretchers for ready use is recommended.

3. When spinal immobilization is required, an approved spinal immobilization device with an appropriately sized collar will be the only devices authorized for transportation and use by RCN Casualty Clearing teams. Once assessed by a Medical Officer or PA, the devices can be removed in a controlled manner and the patient secured in a more comfortable, but controlled position while awaiting evacuation.

4. Hypothermia is a concern in casualties who have suffered trauma. Even in a tropical environment, hypothermia must be planned for and prevented. Blankets or other hypothermia prevention devices are to be used whenever indicated.

5. Annex E and F of this document contain user manuals for both the Ashton Watters Stretcher and the Stokes Litter.

6. Instructions for all the adjunctive equipment used to package patients can be found in Annex G.

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[10] Recommendation #5 Minimal patient handling: a faculty of prehospital care consensus statement. Emerg Med J 2013;30:1065-1066 doi:10.1136/emermed-2013-203205

[11] Minimal patient handling: a faculty of prehospital care consensus statement. Emerg Med J 2013;30:1065-1066 doi:10.1136/emermed-2013-203205

409.2 Modes of transport

1. There are several modes of transport available to the RCN to move casualties. When using them, here are some considerations to be taken into account:

a. Air Transport – The casualty needs to be in a stretcher that is secured to the aircraft with appropriate floatation/buoyancy and in the event of a ditching the aircrew need to know how to rescue the patient from the stretcher. All ship CCT members need to be familiar with how to load and unload the aircraft being used, as well as how to prepare the patient in the stretcher prior to any potential emergency. This shall familiarization shall be conducted whenever the HELO is embarked, and anytime a new aircraft is being used, at the first opportunity. Whenever there are opportunities to practice with different aircraft, the opportunity should be exploited. The following procedures will be used for Ships with hangars and flight decks:

(1) "Hot Loading":

(a) The stretcher will be prepared in the hangar and the casualty secured ready for movement onto the deck with the main hangar door closed;
(b) A team of four CC or Deck Crew will man the stretcher for carrying and be briefed for safety by the Flight Deck Director. They will wear eye, ear and head protection, including the patient;
(c) On command from the Aircraft Commander, the hangar door will be opened high enough to safely exit the hangar and move into the standby position and await permission to approach the designated patient loading door;
(d) The loading door will be approached under the supervision of the Flight Deck Director, and all instructions of the aircrew will be followed; and
(e) Under no circumstances will the loading team move aft of the loading door towards the rear of a helicopter, and the loading team will not leave the casualty/aircraft until the aircraft commander has given permission.

(2) Loading of aircraft not under power:

(a) The above sequence will be followed, or can be modified as directed by the aircraft commander with due attention to safety; and
(b) Particular attention to ensure the patient is wearing the appropriate safety equipment prior to aircraft start up; and

(c) A team of four CC or Deck Crew will man the stretcher for carrying and be briefed for safety by the Flight Deck Director. They will wear eye, ear and head protection, including the patient.

b. Ships Boats – The casualty should be placed in a stretcher (with adequate buoyancy/flotation) into the boat before launching if at all possible, as embarking a stretcher patient into a boat "over the side" is dangerous and difficult at best. The patient must be placed head forward, or in the case of a RHIB may be athwart ships on top of the bench in front of the console. CFCD 105 Chapter 11 provides specific direction to the Deck Department on this topic;

c. Billy Pugh – Used to lower and raise pers from water level to deck level with the crane. Placing patients inside the "circle and netting" provides the safest transition if they are unable to hold onto the outside netting. CFCD 105 Chapter 11 provides specific direction to the Deck Department on this topic;

d. Diver Recovery Position (DRP) – Used to recover a man from the water by attaching the system to a diver/swimmer who jumps into the water. The patient is placed in a "horse collar" under the armpits and is raised vertically. This method is very dangerous for a hypothermic/cold patient. It should only be considered for persons who are not cold and who can be rescued very quickly, such as a witnessed Man Overboard (MOB) in harbour alongside. CFCD 105 Chapter 11 refers; and

e. Life Rafts – It may be necessary to embark casualties into Life Rafts. In this case it is preferable to send stretcher cases down to water level in Ships Boats, by Crane or by Billy Pugh and transfer to Life Boats.

410. Preparing for Casualty Clearing Organisation Team Training (CCOTT)

1. Casualty Clearing Organisation Team Training is conducted annually or whenever there is a significant turnover of crew in the CC organization. Ships should organise their training onboard so that the ships staff level of individual First Aid is up to standard <u>prior to attending the team training</u>. The focus of team training should be to integrate the previously learned individual skills into a collective team training session. Specific emphasis should be on command and control of the team and passing of information and recording data, vice practicing individual First Aid Skills.

2. The team should practice the following skills at a minimum:

a. Plotting a casualty either manually or on BDCS (if vessel equipped);
b. Plotting the team being dispatched;
c. Communicating the discovery of a casualty;
d. Updating command on the initial discovery of a casualty by name;

e. Sending up the CoC, the MIST and initial medical priority of the casualty; and
f. Collecting casualties and preparing them for evacuation or holding until an evacuation can take place.

411. Preparing for Damage Control Organisation Team Training (DCOTT)

1. Damage Control Organisation Team Training is conducted before a WUPS program, bi-annually, or when requested by the unit when there has been significant turnover of ships crew. The focus of the training is on the organizational aspects of the team as a whole, as well as practice of DC skills that are difficult to realistically simulate onboard ships, such as hot Fires and wet floods. Individual First Aid Skills again are not a priority as this is collective team training. All CCT personnel should be used in the training to allow them to consolidate their skills while in the midst of a challenging DC training event.

2. The teams should have practice with at least the following skills:

a. Communicating with command the casualty picture;
b. Establishing alternate communications; and
c. Establishing a firefighter rehabilitation center for a sustained event.

3. While recognized that the CCT will have limitations on access to full BDCS or other communications issues in the DCTF, it is important that they get opportunities to utilize as much as possible and use alternate comms solutions to overcome any limitations in the training scenarios.

412. Use of Memory Aids

1. The use of memory aids by both Medical Care Providers and First Aiders is encouraged at all times. Memory aids enhance medical care and will provide First Aiders who are unsure of what they should be doing with guidance on scene. First Aiders shall have access to a memory aid when responding to casualties. The following subjects at a minimum will be detailed on any memory aid provided:

a. Communication frequencies and nets to be used for CC;
b. Short instruction on how to join whatever nets are required for CC on each type of device onboard;
c. The Mass Casualty Triage Sieve;
d. MIST and report checklists;
e. First Aid general treatment/management flowchart; and
f. Duties and responsibilities as a fire or flood boundary.

2. A sample CCT Aide Memoire is attached as Annex D

413. XO's Situational Report (SITREP)

1. During emergency or action stations, the XO will brief the ship`s company with regular status reports over the ship's main broadcast. The XO will provide advanced notice prior to commencing the sitrep. This report will be in a standard format, and cover aspects of the warfare scenario, the DC picture, out of bounds areas, and will include the number of casualties, and when appropriate, the evacuation plan for the casualties, as this will be all-ship impacting. It is not necessary to mention casualty priorities over the main broadcast (this information is available on the BDCS board), and patient names shall not be used over the main broadcast.

2. Immediately on completion of the XO`s SITREP, all CC I/Cs will either agree with the XO`s SITREP or provide a detailed brief of any inaccuracies over the Med Net. CC HQ will then report to Command through the Command Casualty Communicator.

414. CC Org Locations

1. The spatial separation of CC HQ, Fwd CCT and Aft CCT ensure the survival of capability if damage is taken in the vicinity of the main concentration of the CC Org.

2. SSOs specify the designated primary and secondary locations of CCTs. Each CCT shall at all times be set up to function which shall include communications, POU, manual boards, medical equipment and supplies, and memory aids. It is vital that CCT practice operating not only in their primary but also their secondary locations on a regular basis.

415. Proposed Thought Process for CDC/SMA

1. Some CDC and SMAs find it useful to have a formalized thought process to guide their thinking and actions. The OODA loop provides one process for CDC/SMAs to use in order to absorb and act upon the multitude of stimuli when working in the CC Org.

a. Observe. See, hear, and absorb information about the situation. The CDC and SMA listen to the Medical Net, and are looking at DC boards, and has numerous individuals reporting information directly to him/her. He/she needs to see, hear and absorb DC status, feedback from outstations and CCT on the scene, orders from Command, etc.

b. Orient. Understand and contextualize the information seen and heard. Determine the impact of casualties and DC events, assess whether CC Org is doing the right things in line with the CO's intent. Answer the questions: "What does the new information mean to me?" and "Does this information change the decisions made previously?"

c. Decide. Decide on a course of action. Determine full extent of actions required to accomplish it.

d. Act. Direct the CCT outstations to act upon the decision. Inform Command of changes to the organisation. Eg: relocations, casualty consolidation etc...

Chapter 5 – EMERGENCY PROCEDURES AND PLANS

501. Medical Evaluation of DC teams involved in a fire event

1. In general, all personnel who are involved in a fire onboard RCN Vessels are to be accounted for and observed for ill effects. A log will be made of each person and the vital signs and medical evaluation will be recorded and entered into their medical record. A CF 98 Report on Injuries or Exposure to Toxic Materials will be completed IAW DAOD 5018-2.

2. The CCT will assist in collecting information for the SMA and monitoring vital signs. The following parameters will be recorded at a minimum:

a. Temperature;
b. Pulse;
c. Blood Pressure;
d. Carbon Monoxide level;
e. Mental Status; and
f. Cerebellar Exam.

3. A sample recording log is included as Annex C

502. Sustained Fire/DC Response

1. The detailed vessel specific sustained Fire/DC response plans are outlined in SSOs Vol 2 and the DC Manual (CFCD 132, Chapt 6). Common CC Org issues and best practices include:

a. Prior planning:

(1) Ensure that someone is identified who will set up and organize the rest and recuperation area, when the decision is made to transition to sustained posture. They need to know the cue for them to go to the recuperation area and set up before the whole CC Org Moves;
(2) Prepare a log for recording vital signs (a sample is attached as Annex C);
(3) Consider locations to keep casualties and the rest area. (Should be separated from general view when possible but close together to allow medical staff to watch/roam over both areas); and
(4) Identify special equipment that needs to be in the recuperation area. (CO monitors, Cyanide Poisoning kits, etc.)

b. As soon as a sustained event is identified by command:

(1) Have the pre-identified supervisor and a small team (Usually one CC Team is enough) set up recuperation area before moving the whole team;
(2) Gather equipment that needs to be relocated; and
(3) Deploy the CC Org as indicated in the applicable class specific SSO Vol 2.

c. During the sustained event:

(1) Have someone who can step back and observe to troubleshoot issues with the flow of incoming pers to the recuperation/undressing areas. Pay attention to pers who wander in and out of the recuperation area, while still recovering;
(2) Observe where gas powered compressors are being used to fill bottles during and emergency. Two things of medical concern can occur. One is that the compressor intake and exhaust are improperly placed causing exhaust to contaminate the breathing air, and the second is exhaust from the compressor being blown into the recovery area inadvertently contaminating firefighters with Carbon Monoxide resulting in poor recovery times and a false impression of CO exposure in the fire. On a CPF, setting up the filling stations between the hangar and funnels, or on the flight deck close to the hangar face can cause downdrafts that make these problems worse. The wind and exhaust need to be considered in placement and bottle filling;
(3) Make sure that pers at the scene of the fire are still being tracked and that data is being kept to determine who was where, and when in a fire scene; and
(4) Keep track of pers who do not report to the recuperation area during an event. The Section Bases may have pers who do not report and initial attack teams, containment monitors, on scene commanders and boundaries may be missed.

d. Follow up process:

(1) Pers exposed to the smoke and fire scene need to be evaluated and complete CF98;
(2) Recording of medical notes for all patients encountered in the event needs to be completed; and
(3) Communication with the H Svc CoC needs to be established in order to ensure that all follow up care is initiated and patient tracking systems are put into place.

503. Search and Rescue/Humanitarian Assistance Disaster Relief/Non-Combatant Evacuation Operations (SAR/HADR/NEO)

1. The detailed vessel specific SAR/HADR/NEO response plans are outlined in SSOs Vol 2. Common CC Org issues and best practices include:

a. Prior planning phase:

(1) Identify who will set up and organize the reception area. The reception area will require the admin and supply resources of the Log Dept;
(2) Prepare a log for recording patients/and evacuees; and
(3) All staff need to review the platform specific plans in SSOs Vol 2.

b. Planning Phase:

(1) Focus on preparing the away team for disembarkation should be the first priority if they will be used. In quick reaction scenarios, it is usually the away team prep that slows down the unit level response. The setup of the reception area takes very little time comparatively, and most ships take very little time to accomplish this task;
(2) Although not a primary medical concern, consideration of force protection requirements needs to be appropriate and match the scenario that presents. As a senior leader, if you observe that the response is not proportional to the threat it is your duty to point it out for correction. For example, If you are not prepared to employ deadly force in the reception area (ie: Hangar) then armed guards are not really required there. Potential threats should be cleared at the embarkation point or before they leave the initial contact point, before gathering everyone in a reception area. Unarmed "guides or sentries" or armed guards at access points away from the reception area can be used to control access to the rest of the ship;
(3) Gather equipment that needs to be centralized and either sent to the other vessel/location or sent with the away team; and
(4) Make sure that stretchers and other patient moving devices have the appropriate floatation for transport over water and are readily available for dispatch to the scene when requested.

c. During the event:

(1) Have someone who can step back and observe to troubleshoot issues with the flow of incoming pers;
(2) Make sure that force protection is appropriate for the situation as it develops;

(3) Develop a sustainability plan to allow rotation of pers and "down time" whenever possible; and
(4) Recording of information needs to be clear accurate and on the appropriate forms to allow documentation of the entire event.

d. Follow up process:

(1) Recording of medical notes for all patients encountered in the event needs to be completed; and
(2) Communication with the H Svc CoC needs to be established in order to ensure that all follow up care is initiated and patient tracking systems are put into place.

504. Carbon/Airborne Fiber Precautions (events that generate airborne fiber hazards)

1. Carbon Fiber and Fiber hazard generating materials are becoming more common in structural components of not only aircraft but marine vessels as well. Carbon fibre composite material is composed of very strong, very light finely divided carbon fibres bonded with resins, epoxies and phenols to form an inert and harmless compound. Smoke and fumes from burning composite material are toxic and can cause respiratory illness if inhaled:

a. free carbon fibres released from burned composite material contain absorbed pyrolysis products from the burned chemical binders and as a result, they are assumed to be toxic if inhaled; and
b. in addition, free carbon fibres can cause irritation in the eyes, on the skin and in the respiratory tract and lungs. Fine carbon fibres have sharp ends and edges and can penetrate the skin. Medical assistance may be required for removal of splinters embedded in the skin.

2. All vessels need to consider the problem and be prepared to deal with airborne fiber hazards, whether it be in the form of an aircraft crashed on the deck, responding to a crash scene or recovering material from the water.

3. The minimum amount of protection that is required when exposed to the hazard of airborne fiber is sealed eye protection, N95 mask, and nitrile gloves. Exposed skin can be protected by ensuring that it is covered by existing NCD and flash gear. Exposure to the smoke plume is where the hazard is greatest and the vessel should be positioned to avoid the plume and pers should approach the scene from upwind. Firefighters are fully protected by bunker gear and SCBA. Keeping the scene and contaminated areas wet will minimize the ability of fibers to become airborne until a suitable fixant solution is applied.

4. Decontamination of casualties can be obtained by initially wetting them and removing and sealing exposed clothing in plastic bags whenever possible.

5. The detailed Carbon Fibre Firefighting plan is detailed in the platform specific sections of SSOs Vol 2. Specific CC Org considerations include:

a. Prior planning phase:

(1) Identify the supplies you have onboard to manage airborne fiber hazards;
(2) Prepare a log for recording patients/and evacuees; and
(3) All staff need to review the platform specific plans in SSOs Vol 2.

b. Planning Phase:

(1) Consider how decontamination will be done and how the flow will occur to avoid spreading contamination throughout the ship;
(2) Preposition supplies where everyone knows where to find them; and
(3) Consider the process for transitioning any firefighters from Bunker Gear and SCBA to N95 masks in preparation for decontamination.

c. During the event:

(1) Ensure casualty treatment has priority over contamination. Lifesaving treatment shall not be deferred to prevent spread of contamination. When in doubt, get everything wet and wear minimum protection;
(2) Make sure that personnel who come to the casualty management area are appropriately dressed for the situation as it develops;
(3) Develop a sustainability plan to allow rotation of pers and “down time” whenever possible; and
(4) Recording of information needs to be clear accurate and on the appropriate forms to allow documentation of the entire event.

d. Follow up process:

(1) Recording of medical notes for all patients encountered in the event needs to be completed;
(2) Effectively contain all contaminated materials; and
(3) Communication with the H Svc CoC needs to be established in order to ensure that all follow up care is initiated and patient tracking systems are put into place.

505. Isolated Casualty Response

1. The isolated casualty response is used whenever there are injuries and the casualty clearing organisation has not been stood up by command. There is a slightly different procedure for casualties alongside outside of working hours, but the response for casualties during working hours and at sea are very similar. The vessel specific plan is detailed in applicable SSO Vol 2.

2. The main principles of any casualty response include:

   a. Casualty is discovered, discoverer raises the alarm;
   b. Discoverer provides First Aid/Buddy-Aid to the casualty;
   c. Pre-designated crew respond to the scene (listed in Watch and Station Bill) with first aid equipment;
   d. Lifesaving and stabilization care given;
   e. Casualty handed over to appropriate level of medical care dependant on their condition; and
   f. Shore based Medical Chain of Command needs to be informed to provide follow up and subsequent care.

506. HAZMAT Response

1. The ships HAZMAT response is detailed in CFCD 133 (RCN Emergency Response Team Manual), applicable class Safety and Environmental Management System Manual(SEMS), and SSO Vol 1 and 2.

2. Notwithstanding the above planned responses, the SMA is a valuable resource to the command team and should close up and advise the CSE officer or XO at the scene of any spill and advise with regards to levels of protection required for the substances in the area. This being said, it is incumbent on the SMA and First Responders to know the most dangerous substances onboard their vessel and the actual level of protection required to avoid exposure, and exactly how to rapidly decontaminate an exposed person without having to rely on contacting shore facilities for advice. MSDS Sheets often do not have that much detailed information at hand for decontamination.

3. Medical Teams that may be required to decontaminate must know how they will actually do the decontamination (IE: Soap and water, or just strip and bag?) before they show up on scene. They are not a rapid response element so should not be required to dress without being briefed on what they need to be wearing. Emergencies and casualties in the red zone are handled by pers in SCBA and Bunker Gear.

507. CBRN Casualty response

1. CBRN doctrine for the RCN can be found in the following publications:

   a. C-23-045-000/AG-001 and 002

Damage Control Manual Vols 1 & 2;

b. CFCD 516;

c. B-GJ-005-311/FP-010 CF Nuclear Biological Chemical Defense Operations Chap 3;

d. Allied Tactical Publication 45;

e. Allied Tactical Publication 1(C) Vol. 1 Chap 12;

f. SSOs Vol 1 and 2; and

g. DGI/1200.0000/CA/F/WW/04/94/70494 The CBRN Threat to Canadian Naval Operations dated 28 Sep 94.

2. In addition, CF H Svc Gp has a website of CBRN medical resources at http://cmp-cpm.mil.ca/en/health/personnel-providers/cbrn-medical-resources.page. And NATO Standard AMedP-7.3 Training of Medical Personnel for CBRN Defence provides basic guidance for possible training.

3. Guidance on the specifics of CF H Svc Response to CBRN casualties can also be provided by the Operational Medicine Group at CF H Svc Gp HQ in Ottawa and Canadian Joint Incident Response Unit (CJIRU) through the local Fleet and Formation Surgeons.

4. In general CBRN Medical planning must begin as soon as the threat is assessed and the mission is approved. Some of the decisions that need to be made well in advance of a deployment include:

a. Is there a requirement to embark CBRN Medical Countermeasures (MCM)? OPS staff/CJOC/and Formation Surgeon should provide guidance;

b. Storage of MCM requires a combination of secure spaces and refrigerated spaces. The CPF Sickbay and stores area are not big enough;

c. Is there any extra training required for the medical Staff if any specialized monitoring equipment or testing capability is installed on the ship or utilized to detect the threat; and

d. Are there any briefs or training required for the ships company as a result of embarking Medical Countermeasures (make sure time is reserved in training schedule before ship leaves if possible)? There is usually a general requirement to brief all crew on any medications that may be used that are Surgeon General Controlled Drugs.

5. The CC Org in a CBRN threat environment is essentially collapsed to just the Med Team and perhaps one Casualty Clearing Team if pers are available, as the majority of the Log Dept is used to perform the duties of contamination monitoring and surveys. The Medical Team if embarked will usually be reserved to manage any casualties and augmented with helpers as required in the event of damage or mass casualties.

508. SUBSAR Considerations

1. SUBSAR in the RCN is described in the following publications and policies:

a. The Submarine Rescue Manual (ATP 57);

b. RCN Submarine Escape and Rescue Manual (CFCD 103); and

c. MARPAC OP Order 4/02.

2. Submarine emergencies can and likely will involve every vessel in the area or able to respond. Ships of all classes should be prepared to assist in some capacity to a submarine emergency, from being the ship on the scene and recovering escaping sailors, to the ship that has the specialized responsibility to embark the local SUBSMASH Stores and extra medical/diving resources. Each class of vessel should have a plan to address their potential involvement in a SUBSAR scenario.

3. Particular areas that should be addressed when embarking the SUBSAR Stores are:

a. Having a plan on exactly where each item will be placed and secured when embarked, prior to actually embarking the equipment. The equipment is very heavy and must be secured before proceeding to sea;

b. Having a logical layout of the path that survivors will take in their management through each stage of embarkation, triage and management. There is usually not a lot of room and lots of supplies so efficiently organizing where survivors and treatment teams will be is very important to avoid bottlenecks and moving patients multiple times to gain access to supplies or to

c. Communications – The ships all have many handheld radios with pre-approved frequencies for Damage Control, Force Protection, Boarding Party, that can all be utilized to establish and emergency communication capability for the duration of a SUBSAR Incident. Communication between the embarking specialist teams and the ships staff early on to organize the communications will be exceptionally beneficial. Large ships have an internal telephone system that can be "stood to" with a Medical net that can provide dedicated communications. The BDCS System, and cameras (if installed) can be used to track casualties and assist the Ships Ops staff in developing the evacuation plan, and provide situational awareness to the command teams.

d. The Ships medical staff should engage any embarking Specialist Medical team early on to provide platform specific assistance setting up communications and organizing the team. They should be the resource the SMO turns to, in order to request initial support for the ships staff, such as extra electrical outlets for medical equipment, extra lighting, communications nets that can be established.

It goes without saying that the SMA on the vessel will need to ask for help from the ships staff, but they should know directly who the best resource for each problem on the vessel is.

509. Rescue stations/MOB

1. The class specific medical response to Rescue Stations is described in the applicable class specific SSOs Vol 2.

4. The following equipment at a minimum is to be gathered, checked for serviceability, and set up at the triage area (on a CPF in the Hangar, MCDV in the wardroom) while waiting for the recovery of the Man Overboard. The SMA is responsible for ensuring that this is done, prior to attending the main site of recovery (Boat deck on CPF, Sweep Deck of MCDV):

a. Oxygen Delivery system;
b. AED;
c. Patient warming equipment (class specific);
d. Dry Blankets and sheets; and
e. First Aid equipment/Bags.

5. The SMA will ensure that the Casualty is triaged quickly on the boat deck and moved to a protected location as efficiently as possible to begin the warming process and management of any injuries sustained.

6. At all times the casualty is to be handled as gently as possible to avoid cardiac irritation and subsequent cardiac arrhythmia. The casualty shall be removed from wet clothing and equipment and placed into a dry warm environment as soon as practical. Rapid rewarming in a shower of a cold casualty should be avoided and can be dangerous in very cold casualties.

510. Emergency Flying Stations (EFS)

1. The class specific medical response to EFS is described in the applicable class specific SSOs Vol 2.

2. The general response to EFS requires the medical team to consider the following in their response:

a. It is possible that the aircrew or deck pers involved in an emergency may fall overboard during the recovery. All pers must be accurately accounted for in any Crash on Deck scenario. Bringing the ship to Verification Muster during an emergency would not be possible, so exact personnel accountability is key to avoid confusion;

b. A crash on deck is always possible when the ship is at normal flying stations, so the response to such emergencies should be practiced as part of the normal ships operations (vice always practicing as part of a EFS Drill); and

c. As part of the EFS drill, it is important that once everyone is mustered, that they rehearse their responsibilities and actions on the pipe "Crash on Deck" before it is made, and are ready to take action immediately. They need to be paired up with their partner, have their first aid equipment in their possession and know their route to the scene.

511. Hostile and Violent patients

1. The medical management of Hostile and Violent Patients is generally defined and specified in CF H Svcs Gp Instruction 4090-13.

2. Each vessel is to conduct regular training in managing hostile and violent patients.

3. The outline of the expected premise to be followed by ships staff is the following:

a. Alongside:

(1) On discovering a hostile or violent member of the crew, the OOD is to be piped to the scene and will assume control of the situation;

(2) The OOD is to exercise judgement in de-escalating the situation using the minimum force necessary to ensure the safety of the person and the persons assisting them. The OOD primary responsibility is to focus on de-escalating and scene safety of the situation. And all measures including talking to the person should be exhausted before force is used. If isolating the person in a safe place is possible, then doing so while assembling help to get the situation under control is preferable to using immediate direct force to subdue a violent patient;

(3) The OOD should consider using the following in their de-escalation attempts:

- Verbally requesting cooperation while maintaining a nonaggressive posture and tone of voice;
- Having an adequate security force nearby that is visible to the patient;
- Redirecting and/or diverting the patient's emotions; and
- Separating the patient from others.

(4) Keeping the above in mind, there are certain indicators that should be watched for which may indicate that physical restraint will be imminently necessary:

- When the patient is physically combative;
- When the patient is a clear and immediate danger to self or others;
- When less restrictive alternatives have been attempted without success; and
- When it reasonably appears that delay in restraint would subject the patient and others to risk of serious harm.

(5) Concurrently, the DCOXN is to prepare a team to execute restraint should it be required. The DCOXN is not to forcibly restrain a hostile or violent member of the crew until a team of no less than five members is present (one per limb and one to control the head);

(6) When at all possible, hard restraints such as handcuffs or "Zip Ties" are not to be used when the patient is still struggling. In all cases where hard restraints are used the patient will be continuously monitored at all times for subsequent injury from the restraints, and segregated from the rest of the crew in a designated quiet area;

(7) In cases where physical restraint is required to subdue a violent patient, consideration should be given to actually performing an arrest of the individual and consulting command and legal advice once the situation is brought under control;

(8) All physically restrained personnel are to be immediately referred to medical care for examination and evaluation; and

(9) The command team is to be immediately notified by the OOD either by recall or overhead broadcast at the earliest opportunity of an incident where force has been used on a hostile individual onboard the vessel.

b. At Sea:

(1) All the above are to be followed with the exception that the OOD duties are performed by the XO and the DCOXN duties are to be performed by the COXN.

511. Death of Casualties

1. Casualties that expire as a result of their injuries are to be handled IAW current CAF Policies. Maintaining the respect and dignity of all deceased pers will be the main focus of everyone involved in the handling of human remains.

4. The following considerations shall be observed:

a. Any interventions that have been attempted are to remain in place. No attempt to remove IV's, bandages, or other tubes are to be made. Any bandages or consumable items are to be secured and left with the body;

b. The ships process for evidence gathering shall commence as soon as possible and if the body needs to be moved photos shall be taken before that occurs;
c. The only photos that are to be taken are for evidence gathering purposes;
d. The body shall be covered and protected from the elements at all times when the evidence gathering is able to accommodate;
e. Before the body is moved the applicable Fleet support MO and Coroner will be contacted for advice with regards to appropriate storage; and
f. Human remains shall not be stored in refrigerators or where food is normally stored. Potato lockers, fridges and freezers are not to be used to store human remains.

ANNEX A – CCT EQUIPMENT AND SUPPLY REQUIREMENTS

1. Items mentioned in this Annex are to be considered as a minimum recommendation. Units are responsible to maintain emergency equipment, have it readily available in the event of action or emergency situations. These items shall be located in the vicinity of CC Org locations in order to be accessed by the team as they respond to a casualty. The contents of kits may be added to or altered to be relevant for the class and mission of the ship. Applicable class SSOs in Vol 2 will define specific CC Org requirements.

2. CC HQ shall be fitted with the following equipment:

   a. portable manual plotting board;

   b. POU/BDCS (if installed in the class of vessel);

   c. internal communications using PACE; and

   d. emergency lighting.

3. Each CCT Outstation shall be fitted with the following equipment:

   a. portable manual plotting board;

   b. POU/BDCS (if installed in the class of vessel);

   c. internal communications using PACE;

   d. emergency lighting; and

   e. Medical Equipment consisting of:

      (1) Oxygen delivery system with spare tanks;
      (2) Medical supplies in personal bags, as well as resupply items contained in storage boxes;
      (3) AED;
      (4) Monitor for Patient vital signs;
      (5) Traction Splints; and
      (6) Access to Stretchers.

NOTE

Equipment is to be checked during Pre-Sail checks as per SSOs and is to include setting up BDCS and the portable Manual board IAW the current Watch and Station Bill.

ANNEX B – SMART TRIAGE CARD SAMPLE

Casualty Tags - Blank forms for trainir

ANNEX C – FIREFIGHTER LOG

Sustained Fire Med Log.xlsx

ANNEX D – FIRST AIDER AIDE MEMOIRE

RCN CCT Aide Memoire.docx

ANNEX E – ASHTON WATERS USER GUIDE

Ashton Waters Manual English.pdf

Ashtome Waters Manual Francais.pdf

ANNEX F – STOKES LITTER USER GUIDE

Stokes Litter Manual.pdf

ANNEX G – SPECIALIZED EQUIPMENT USER GUIDES

KED User Manual.pdf

Traction Splint Cbt page 1.jpg

Traction Splint Cbt Page 2.jpg

CAT Tourniquet Version 7 Manual.pdf